MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, BINARY, DIGITAL, KEY CODE ZBY

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic binary counter module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.147 failures per million operating hours for class I or 0.469 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be ZBY.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/46-1 for class I and M28787/46-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

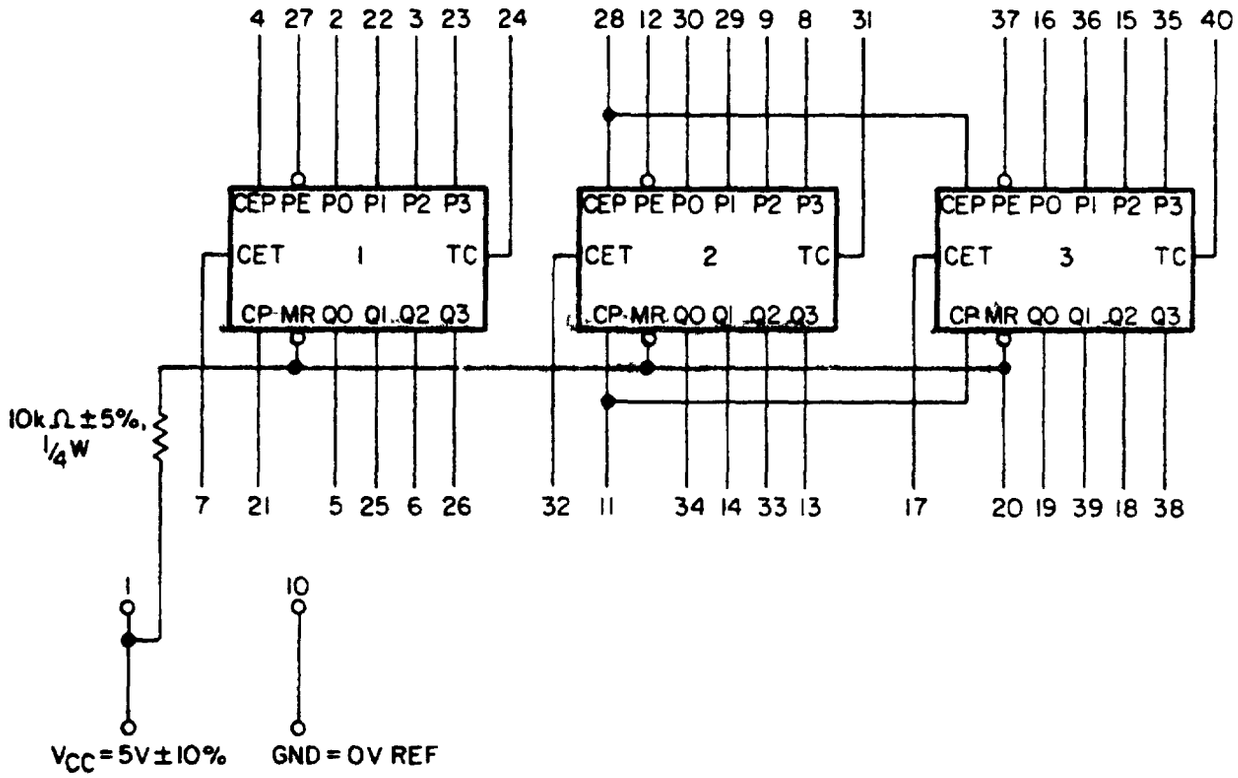
4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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NOTES
1. P0, P1, P2, P3 = Parallel inputs.
2 Q0, Q1, Q2, Q3 = Parallel outputs
3 TC = Terminal count.
4 PE = Parallel enable input
5 CP = Clock input.
6. CEP = Count enable parallel input.
7. CET = Count enable trickle input.
8. MR = Master reset input.

FIGURE 1. ZBY module function diagram, binary counter.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A15	V_{OUT} (low-loaded)	IV	S	10	600	10	600	10	600	10	600	10	600	mV
B1-B20	Input breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
B21,B22	Input breakdown current	V	S	0	2	0	2	0	2	0	2	0	2	mA
B23	Input breakdown current	V	S	0	3	0	3	0	3	0	3	0	3	mA
C1-C23	Input clamp diode voltage	VI	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D	Isolation													
	Initial	VII	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	VII	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
E1-E3	Functional	VIII	X	← See table VIII →										
F1-F9	Functional	VIII	X	← See table VIII →										
G1-G15	Output voltage, high level	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
H1-H15	Output voltage, low level	IX	X	0	300	0	300	0	300	0	300	0	300	mV
J1-J3	Input threshold low level check	X	X	← See table X →										
K1-K6	Input threshold high level check	X	X	← See table X →										
L1	Input current, low level	XII	X	0.82	1.88	0.82	1.88	0.82	1.88	0.82	1.88	0.82	1.88	mA
L2-L9	Input current, low level	XII	X	0.10	0.86	0.10	0.86	0.10	0.86	0.10	0.86	0.10	0.86	mA
L10	Input current, low level	XII	X	0.10	1.72	0.10	1.72	0.10	1.72	0.10	1.72	0.10	1.72	mA
L11	Input current, low level	XII	X	0.10	0.43	0.10	0.43	0.10	0.43	0.10	0.43	0.10	0.43	mA
L12-L23	Input current, low level	XII	X	0.10	0.29	0.10	0.29	0.10	0.29	0.10	0.29	0.10	0.29	mA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1	Input current, high level	XII	X	-228	-336	-228	-336	-228	-336	-228	-336	-228	-336	µA
M2-M9	Input current, high level	XII	X	0	40	0	40	0	40	0	40	0	40	µA
M10	Input current, high level	XII	X	0	80	0	80	0	80	0	80	0	80	µA
M11	Input current, high level	XII	X	0	20	0	20	0	20	0	20	0	20	µA
M12-M23	Input current, high level	XII	X	0.0	13.3	0.0	13.3	0.0	13.3	0.0	13.3	0.0	13.3	µA
N1-N15	Output short circuit current	XIII	X	2.5	25.0	2.5	25.0	2.5	25.0	2.5	25.0	2.5	25.0	mA
P1,P2	Power supply current	XIV	S	1.0	82.5	1.0	82.5	1.0	82.5	1.0	82.5	1.0	82.5	mA
Q1	Input capacitance	XV	Q	3	45	3	45	3	45	3	45	3	45	pF
Q2,Q3	Input capacitance	XV	Q	2	30	2	30	2	30	2	30	2	30	pF
Q4-Q23	Input capacitance	XV	Q	2	15	2	15	2	15	2	15	2	15	pF
R	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
S1-S3	Propagation delay time (t_{PHL}), C_L = 40 pF	XVII	S	10	49	10	52	10	52	10	52	10	52	ns
S4-S15	Propagation delay time (t_{PHL}), C_L = 40 pF	XVII	S	15	59	15	62	15	62	15	62	15	62	ns
T1-T3	Propagation delay time (t_{PHL}), C_L = 100 pF	XVII	Q	10	55	10	58	10	58	10	58	10	58	ns
T4-T15	Propagation delay time (t_{PHL}), C_L = 100 pF	XVII	Q	15	65	15	68	15	68	15	68	15	68	ns
U1-U15	Transition time (t_{THL}), C_L = 40 pF	XVIII	S	2	20	2	22	2	22	2	22	2	22	ns
V1-V15	Transition time (t_{THL}), C_L = 100 pF	XVIII	Q	2	25	2	27	2	27	2	27	2	27	ns

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
W1-W3	Propagation delay time (t_{PLH}), C_L = 40 pF	XIX	S	20	99	20	105	20	105	20	105	20	105	ns
W4-W15	Propagation delay time (t_{PLH}), C_L = 40 pF	XIX	S	10	54	10	59	10	59	10	59	10	59	ns
X1-X3	Propagation delay time (t_{PLH}), C_L = 100 pF	XIX	Q	25	105	25	111	25	111	25	111	25	111	ns
X4-X15	Propagation delay time (t_{PLH}), C_L = 100 pF	XIX	Q	15	60	15	65	15	65	15	65	15	65	ns
Y1-Y15	Transition time (t_{TLH}), C_L = 40 pF	XX	S	5	45	5	50	5	50	5	50	5	50	ns
Z1-Z15	Transition time (t_{TLH}), C_L = 100 pF	XX	Q	10	55	10	60	10	60	10	60	10	60	ns
AA1-AA15	Propagation delay time (t_{PHL}) and minimum pulse width and recovery time check, C_L = 40 pF	XXI	S	25	99	25	105	25	105	25	105	25	105	ns
BB1-BB15	Propagation delay time (t_{PHL}) and minimum pulse width and recovery time check, C_L = 100 pF	XXI	Q	25	105	25	111	25	111	25	111	25	111	ns
CC1-CC3	Propagation delay time (t_{PHL}), C_L = 40 pF	XXII	S	10	64	10	68	10	68	10	68	10	68	ns
CC4-CC6	Propagation delay time (t_{PLH}), C_L = 40 pF	XXII	S	10	59	10	63	10	63	10	63	10	63	ns
DD1-DD3	Propagation delay time (t_{PHL}), C_L = 100 pF	XXII	Q	15	70	15	75	15	75	15	75	15	75	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
DD4-DD6	Propagation delay time (t_{PLH}), C_L = 100 pF	XXII	Q	15	65	15	70	15	70	15	70	15	70	ns
EE1-EE6	Maximum counting frequency (minimum clock pulse width check)	XXIII	S	0.97	1.07	0.97	1.07	0.97	1.07	0.97	1.07	0.97	1.07	µs
FF1-FF6	Setup and hold time (measuring output period) check	XXIV	S	3.0	3.4	3.0	3.4	3.0	3.4	3.0	3.4	3.0	3.4	µs
GG1-GG12	Setup and hold time (measuring T_{PLH}), P high, check	XXV	S	10	100	10	105	10	105	10	105	10	105	ns
GG13-GG15	Setup and hold time (measuring t_{PLH}), PE, check	XXV	S	10	100	10	105	10	105	10	105	10	105	ns
GG16-GG27	Setup and hold time (measuring t_{PHL}), P low, check	XXV	S	10	100	10	105	10	105	10	105	10	105	ns
HH1-HH15	Output crosstalk, high level	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
JJ1-JJ15	Output crosstalk, low level	XXVI	Q	0	800	0	800	0	800	0	800	0	800	mV
JJ16-JJ30	Output crosstalk, low level, duration	XXVI	Q	0	400	0	400	0	400	0	400	0	400	mV
KK	Power supply filter capacitance	XXVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through KK shall be performed in any order with the following exceptions. If test conditions A, B, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through KK in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C, E through P, and R through KK shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition KK, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C, E through P, and R through KK shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C and E through KK shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Part number M28787/46-1 supersedes part number M28787/46. The two parts are interchangeable.

* 6.5 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

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		Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)		455	455	mW
Supply voltage		5.0 ±10%	5.0 ±10%	V dc
Clock input (C_L = 40 pF)				
Frequency		8 max	8 max	MHz
Pulse width (high level)		50 min	50 min	ns
Trigger edge		Positive	Positive	---
Input voltage				
Low level		-0.5 min, 0.7 max	-0.5 min, 0.7 max	V
High level		2.0 min, V_{CC} max	2.0 min, V_{CC} max	V
Input current, low level				
P0, P1, P2, P3	(V_{IL} = 0.0 V)	-0.1 min, -0.29 max	-0.1 min, -0.29 max	mA
	(V_{IL} = 0.3 V)	-0.1 min, -0.27 max	-0.1 min, -0.27 max	mA
1CEP	(V_{IL} = 0.0 V)	-0.1 min, -0.43 max	-0.1 min, -0.43 max	mA
	(V_{IL} = 0.3 V)	-0.1 min, -0.40 max	-0.1 min, -0.40 max	mA
2-3CP	(V_{IL} = 0.0 V)	-0.1 min, -1.72 max	-0.1 min, -1.72 max	mA
	(V_{IL} = 0.3 V)	-0.1 min, -1.60 max	-0.1 min, -1.60 max	mA
1CP, CET, PE	(V_{IL} = 0.0 V)	-0.1 min, -0.86 max	-0.1 min, -0.86 max	mA
	(V_{IL} = 0.3 V)	-0.1 min, -0.80 max	-0.1 min, -0.80 max	mA
2-3CEP	(V_{IL} = 0.0 V)	-0.1 min, -0.86 max	-0.1 min, -0.86 max	mA
	(V_{IL} = 0.3 V)	-0.1 min, -0.80 max	-0.1 min, -0.80 max	mA
MR	(V_{IL} = 0.0 V)	-0.82 min, -1.88 max	-0.82 min, -1.88 max	mA
	(V_{IL} = 0.3 V)	-1.75 max	-1.75 max	mA
Input current, high level				
P0, P1, P2, P3	(V_{IH} = 2.4 V)	0 min, 13.3 max	0 min, 13.3 max	µA
	(V_{IH} = 5.5 V)	0 min, 1.0 max	0 min, 1.0 max	mA
1CEP	(V_{IH} = 2.4 V)	0 min, 20 max	0 min, 20 max	µA
	(V_{IH} = 5.5 V)	0 min, 1.0 max	0 min, 1.0 max	mA
2-3CP	(V_{IH} = 2.4 V)	0 min, 80 max	0 min, 80 max	µA
	(V_{IH} = 5.5 V)	0 min, 2.0 max	0 min, 2.0 max	mA
1CP, CET, PE	(V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
	(V_{IH} = 5.5 V)	0 min, 1.0 max	0 min, 1.0 max	mA
2-3CEP	(V_{IH} = 2.4 V)	0 min, 40 max	0 min, 40 max	µA
	(V_{IH} = 5.5 V)	0 min, 2.0 max	0 min, 2.0 max	mA
MR	(V_{IH} = 2.4 V)	-0.228 min, -0.336 max	-0.228 min, -0.336 max	mA
	(V_{IH} = 5.5 V)	3.0 max	3.0 max	mA
Output voltage				
Low level		0.0 min, 0.3 max	0.0 min, 0.3 max	V
High level		2.4 min, V_{CC} max	2.4 min, V_{CC} max	V
Output drive current maximum capability				
Low level	(V_{OL} = 0.3 V)	3.2 max	3.2 max	mA
High level	(V_{OH} = 2.4 V)	-0.32 max	-0.32 max	mA
Setup time				
CET, CEP		80 min	80 min	ns
P0, P1, P2, P3		110 min	110 min	ns
PE		80 min	80 min	ns
Hold time				
CET, CEP		0 min	0 min	ns
P0, P1, P2, P3		0 min	0 min	ns
PE		0 min	0 min	ns
Reset pulse width		65 min	65 min	ns
Reset recovery time		75 min	75 min	ns

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* 6.6 Module function. The module contains three 4-bit synchronous binary counters with independent inputs and outputs with the exception that circuits 2 and 3 have common clocks (CP) and count enable parallel inputs (CEP). All counters can be reset with a master reset (MR).

6.6.1 Input to output relationship. Each binary counter has the input to output relationship of table II for positive logic.

6.6.1.1 Functional description. The module contains three binary counters which are fully synchronous with the clock pulse. Each counter contains four master-slave flip-flops which are driven, in parallel, by the clock through a clock buffer. The master is inhibited from further change during the low-level to high-level transition of the clock. Data is transferred from the masters to the slaves and reflected at the outputs after the masters are inhibited. When the clock is high the masters remain inhibited and the master-slave data path remains established. During the clock high-to-low transition, the slave is inhibited from further change, followed by the enabling of the masters for acceptance of new data from the counting logic or the parallel entry logic.

6.6.1.2 Control inputs. There are three control inputs: Parallel enable (PE), count enable parallel (CEP), and count enable trickle (CET). These control inputs select the mode of operation as shown in table III. When the conditions for counting are satisfied, that is, CEP, CET, and PE are high, the rising edge of the clock will change the counters to the state of the count sequence.

6.6.1.3 Counter setting. The counter can be synchronously preset from the four parallel inputs (P0-P3) when PE is low. When the PE and clock are low, each master of the flip-flops is connected to the appropriate parallel input (P0-P3) and the slaves (outputs) are steady in their previous state. When the clock goes high, this information is transferred to the slaves and reflected at the outputs. The PE overrides both count enables presetting the counter when low.

6.6.1.4 State of terminal count. Terminal count is high when the counter is at a terminal count (state 15) and the CET is high.

6.6.1.5 Overriding provisions. The asynchronous master reset (MR) common to all counters, overrides all other inputs resetting each output low.

6.6.1.6 Special conditions. If PE goes high while the clock is low and count enables are not active (low) during the low portion of the clock, the low-to-high clock transition will change Q0-Q3 to the P0-P3 data that existed at the setup time before the rising edge of PE.

a. If the count enables are active (high) during some portion of the clock low period the PE is high during the entire clock low period and the subsequent low-to-high transition will change Q0-Q3 to the next count value.

b. If PE goes high while the clock is low but the count enables are active (high) during some portion of the remaining clock low period, the counter will perform a mixture of counting and loading. On the low-to-high clock transition outputs Q0-Q3 will change as the count sequence or the loading requires. Only the outputs that would not change in the count sequence, and are also reloaded with their present value, stay constant.

6.6.1.7 Common CEP and master reset. Two of the counters in the module have common CEP and all three have a common master reset.

6.6.1.8 High-speed look-ahead technique. Multistage synchronous counting at high speeds is made possible with a high-speed look-ahead technique utilizing the count enables and terminal count logic.

6.6.1.9 Terminal count. TC = CET·Q0·Q1·Q2·Q3. The TC output is asynchronous, therefore, the TC output shoud never be used outside the counting circuit as a synchronous output unless resynchronizing circuitry is added to compensate for the delay differences between the TC and Q outputs.

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When connected as a multistage counter (TC output connected to CET input), the delay t_{PHL} from CP to TC of a counter may exceed the dealy t_{PLH} from CP to QC of the following (adjacent) counter. Under the conditions of the first circuit changing from count 15_{10} to 0 and the following (MSB) circuit changing from count 14_{10} to 15_{10}, the TC output of that MSB counter could generate an illegal logic "H" pulse

6.7 Parallel outputs. Paralleling of outputs is not permissible

6.8 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing activity
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TABLE II. Input to output relationship. 1/-10/

Control inputs		Preset inputs					Reset input	Clock input	Outputs				
CET	CEP	PE	P3	P2	P1	P0	MR	CP	TC	Q3	Q2	Q1	Q0
Reset operation													
X	X	X	X	X	X	X	L	X	L	L	L	L	L
Preset operation													
L	L	H	X	X	X	X	H	L→H	NC	NC	NC	NC	NC
L	H	H	X	X	X	X	H	L→H	NC	NC	NC	NC	NC
H	L	H	X	X	X	X	H	L→H	NC	NC	NC	NC	NC
X	X	L	D3	D2	D1	D0	H	L→H	X	D3	D2	D1	D0
L	X	L	D3	D2	D1	D0	H	L→H	L	D3	D2	D1	D0
H	X	L	H	H	H	H	H	L→H	H	H	H	H	H
H	X	L	L	L	L	L	H	L→H	L	L	L	L	L
Counting operation													
H	H	H	X	X	X	X	H	L→H	L	L	L	L	H
H	H	H	X	X	X	X	H	L→H	L	L	L	H	L
H	H	H	X	X	X	X	H	L→H	L	L	L	H	H
H	H	H	X	X	X	X	H	L→H	L	L	H	L	L
H	H	H	X	X	X	X	H	L→H	L	L	H	L	H
H	H	H	X	X	X	X	H	L→H	L	L	H	H	L
H	H	H	X	X	X	X	H	L→H	L	L	H	H	H
H	H	H	X	X	X	X	H	L→H	L	H	L	L	L
H	H	H	X	X	X	X	H	L→H	L	H	L	L	H
H	H	H	X	X	X	X	H	L→H	L	H	L	H	L
H	H	H	X	X	X	X	H	L→H	L	H	L	H	H
H	H	H	X	X	X	X	H	L→H	L	H	H	L	L
H	H	H	X	X	X	X	H	L→H	L	H	H	L	H
H	H	H	X	X	X	X	H	L→H	L	H	H	H	L
H	H	H	X	X	X	X	H	L→H	H	H	H	H	H
H	H	H	X	X	X	X	H	L→H	L	L	L	L	L

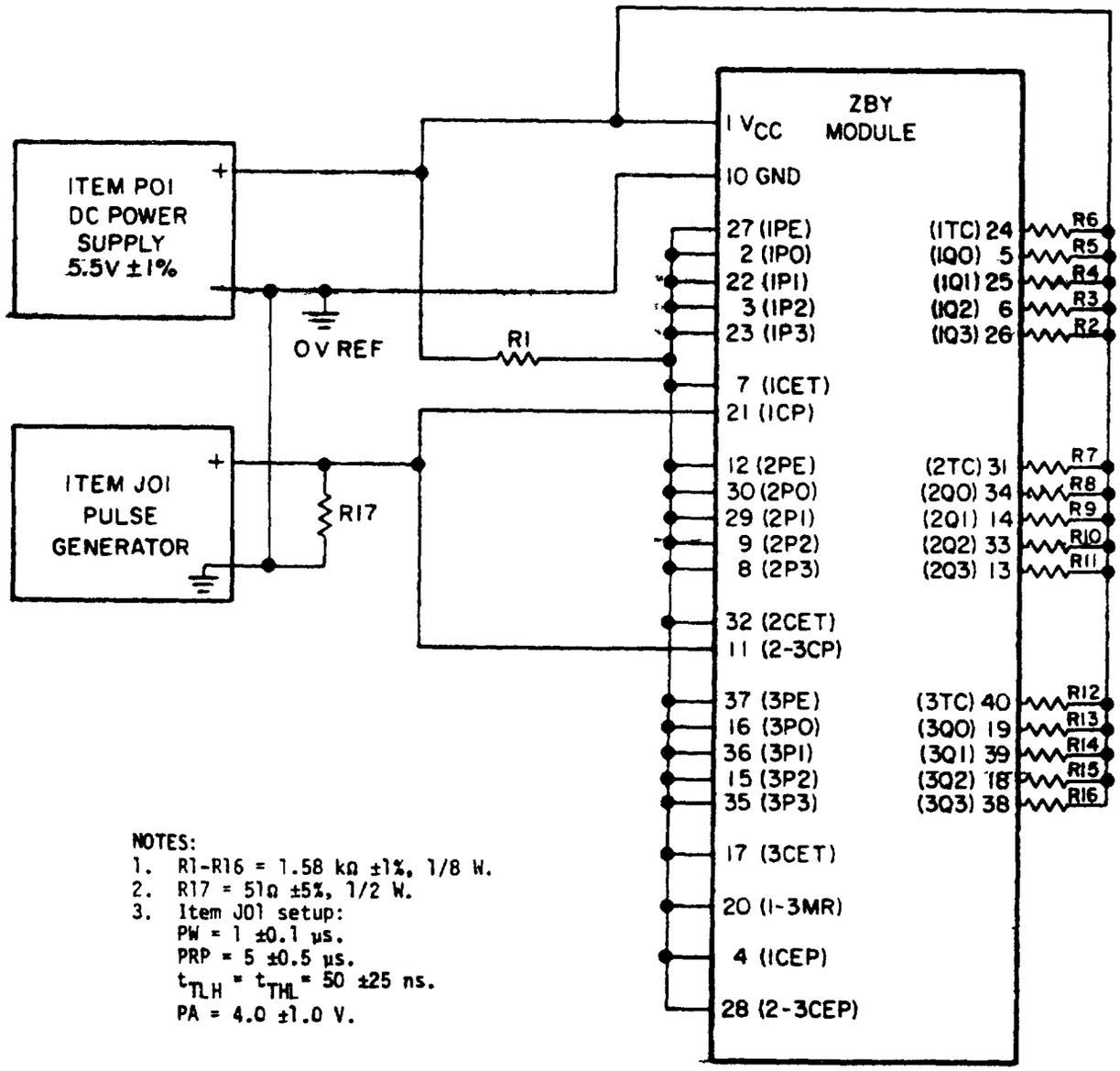
1/ H = High level.
2/ L = Low level.
3/ L→H = Low-level to high-level transition of clock pulse.
4/ NC = No change from previous level.
5/ X = Unspecified inputs or outputs (either high level or low level).
6/ D0-D3 = Known data (either high level or low level).
7/ Counting operation starts at previous output conditions.
8/ TC (terminal count) = CET · Q3 · Q2 · Q1 · Q0.
9/ Reset operation overrides all other operations.
10/ If D0-D3 are "H", then TC is "H", otherwise TC is "L".

TABLE III. Mode selection relationship. 1/

PE	CEP	CET	Mode
L	L	L	Preset
L	L	H	Preset
L	H	L	Preset
L	H	H	Preset
H	L	L	No change
H	L	H	No change
H	H	L	No change
H	H	H	Count

1/ MR = High.

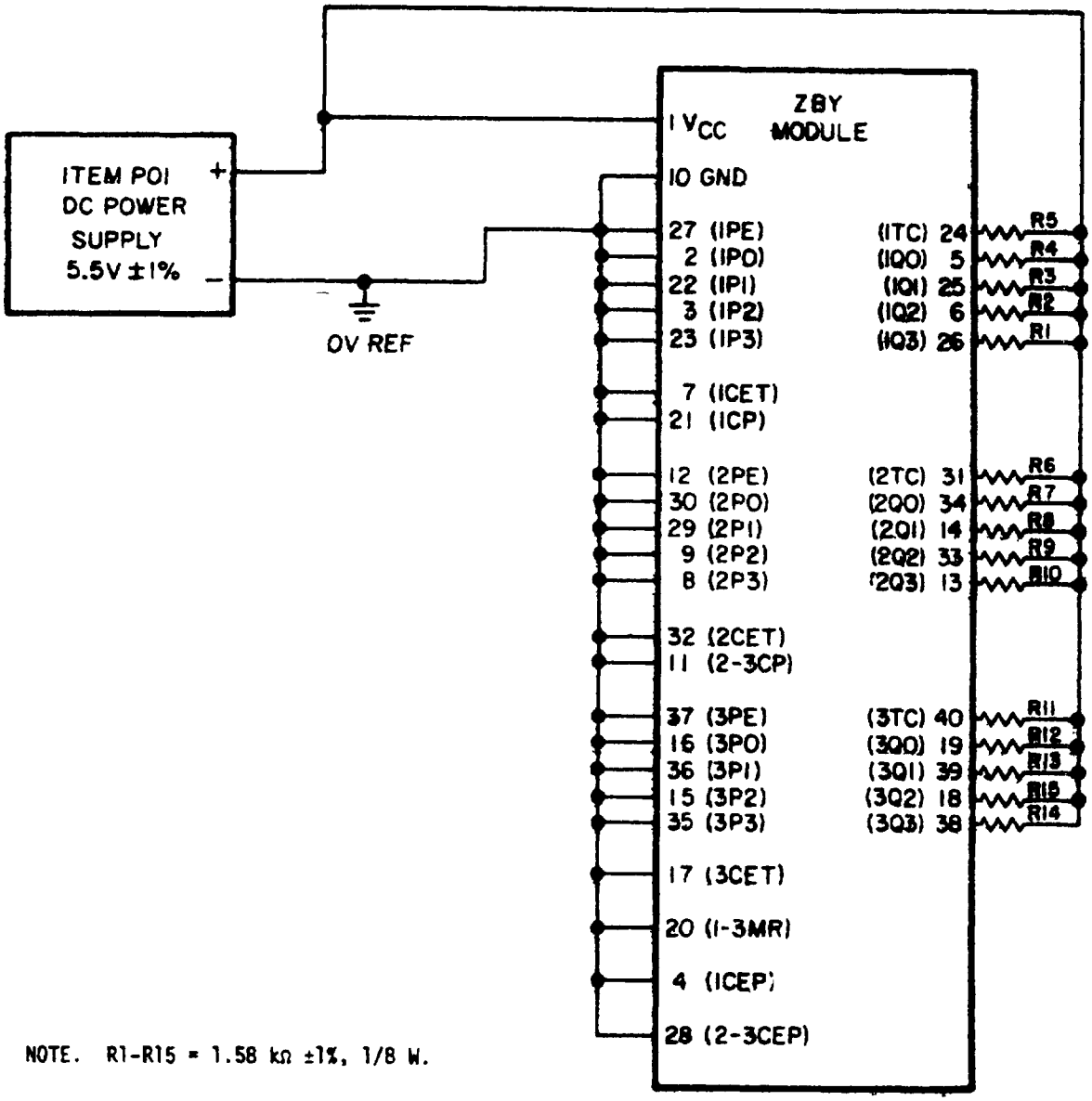
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NOTES:
1. R1-R16 = 1.58 kΩ ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1/2 W.
3. Item J01 setup:
   PW = 1 ±0.1 µs.
   PRP = 5 ±0.5 µs.
   $t_{TLH} = t_{THL} = 50 \pm 25$ ns.
   PA = 4.0 ±1.0 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTE. R1-R15 = 1.58 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit

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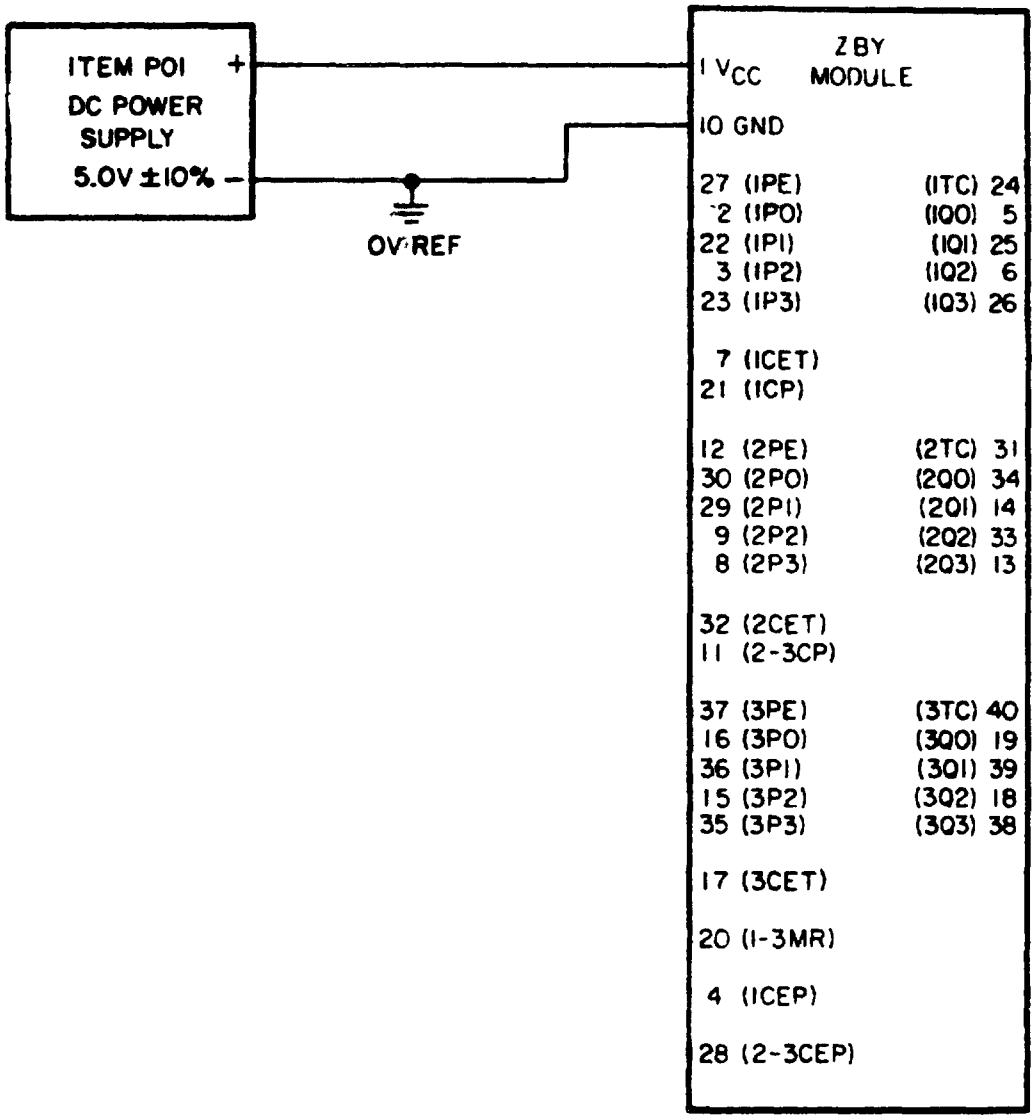


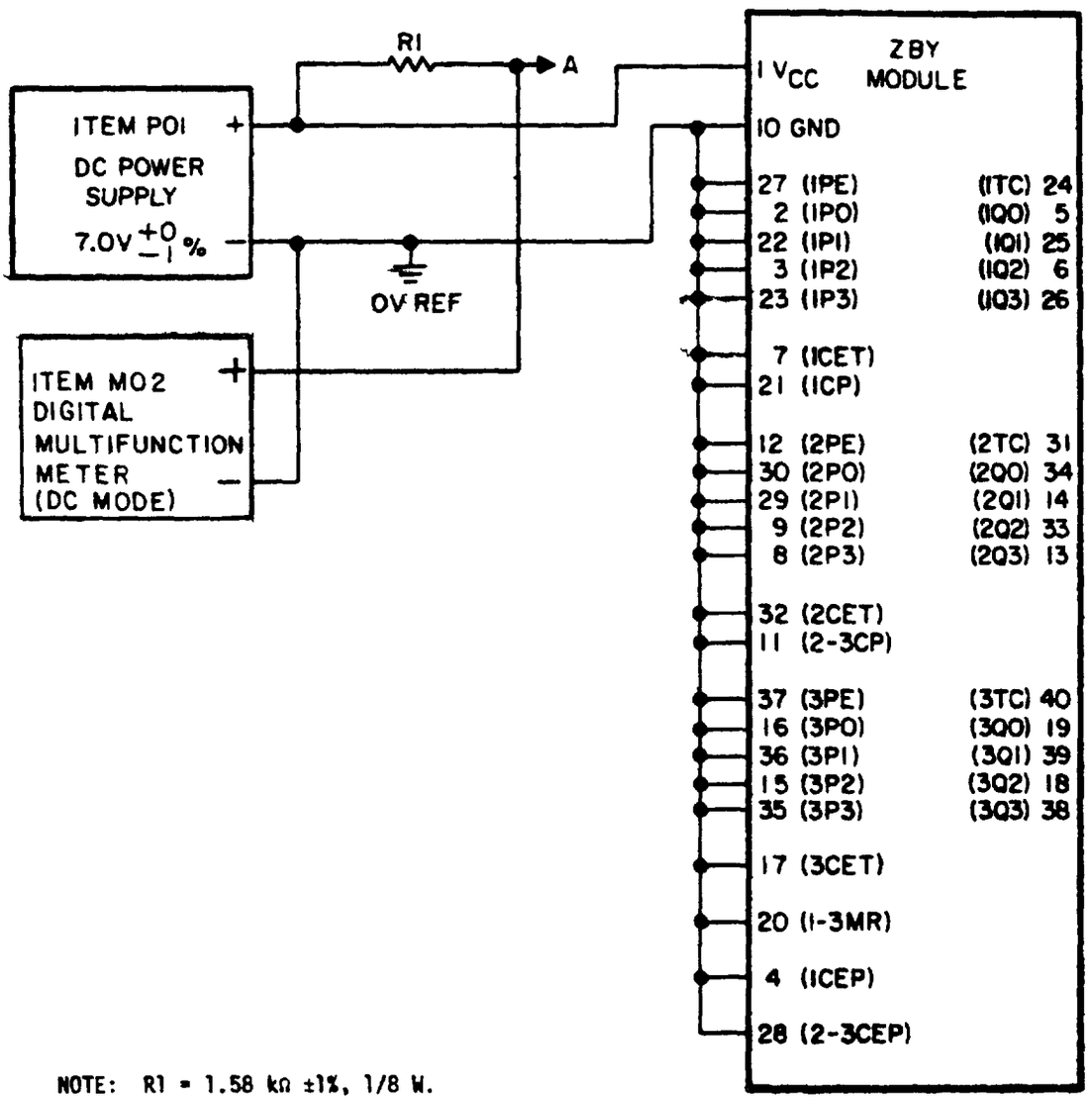
FIGURE 4. Vibration test circuit.

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* TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
A1	1Q3 V_{OUT} (low loaded)	26	10	600	mV	5
A2	1Q2	6				
A3	1Q1	25				
A4	1Q0	5				
A5	1TC	24				
A6	2Q3	13				
A7	2Q2	33				
A8	2Q1	14				
A9	2Q0	34				
A10	2TC	31				
A11	3Q3	38				
A12	3Q2	18				
A13	3Q1	39				
A14	3Q0	19				
A15	3TC	40				

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NOTE: R1 = 1.58 kΩ ±1%, 1/8 W.

FIGURE 5. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

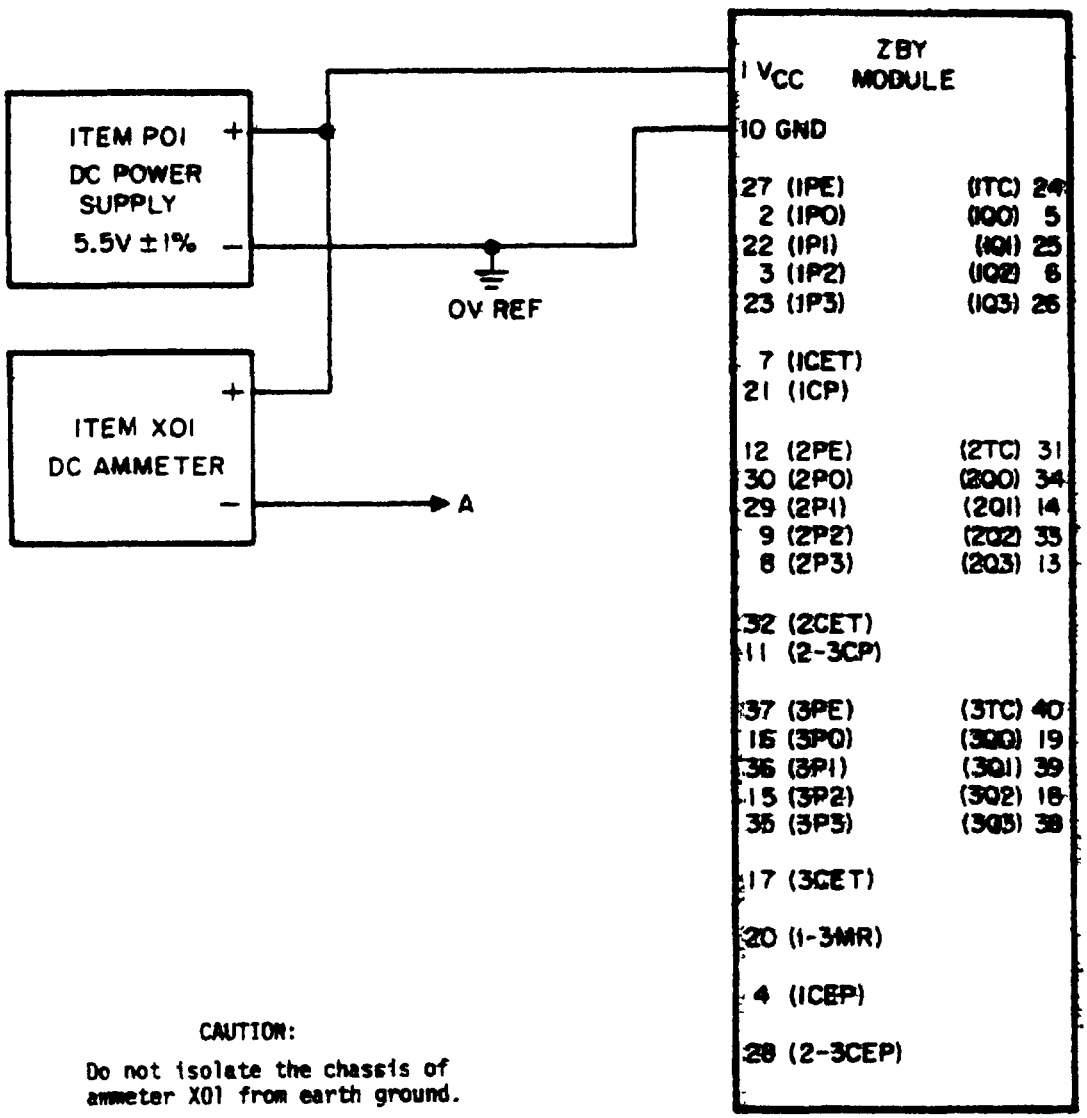
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Pin 10 to pin	Min	Max		
B1	1CEP Input breakdown current	4	2,3,7,20,21,22,23,27	0	1	mA	6
B2	1CP	21	2,3,4,7,20,22,23,27				
B3	1CET	7	2,3,4,20,21,22,23,27				
B4	1PE	27	2,3,4,7,20,21,22,23				
B5	2PE	12	8,9,11,20,28,29,30,32				
B6	2CET	32	8,9,11,12,20,28,29,30				
B7	3CET	17	11,15,16,20,28,35,36,37				
B8	1PO	2	3,4,7,20,21,22,23				
B9	1P1	22	2,3,4,7,20,21,23				
B10	1P2	3	2,4,7,20,21,22,23				
B11	1P3	23	2,3,4,7,20,21,22				
B12	2PO	30	8,9,11,20,28,29,32				
B13	2P1	29	8,9,11,20,28,30,32				
B14	2P2	9	8,11,20,28,29,30,32				
B15	2P3	8	9,11,20,28,29,30,32				
B16	3P3	35	11,15,16,17,20,28,36				
B17	3P2	15	11,16,17,20,28,35,36				
B18	3P1	36	11,15,16,17,20,28,35				
B19	3PO	16	11,15,17,20,28,35,36				
B20	3PE	37	11,15,16,17,20,28,35,36				

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Pin 10 to pin	Min	Max		
B21	2-3CEP Input breakdown current	28	8,9,11,12,15,16,17, 20,29,30,32,35,36,37	0	2	mA	6
B22	2-3CP Input breakdown current	11	8,9,12,15,16,17,20, 28,29,30,32,35,36,37	0	2	mA	6
B23	1-3MR Input breakdown current	20	2,3,4,7,8,9,11,12,15, 16,17,21,22,23,27,28, 29,30,32,35,36,37	0	3	mA	6

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 6. Input breakdown current test circuit for table V.

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TABLE VI. Input clamp diode voltage test.

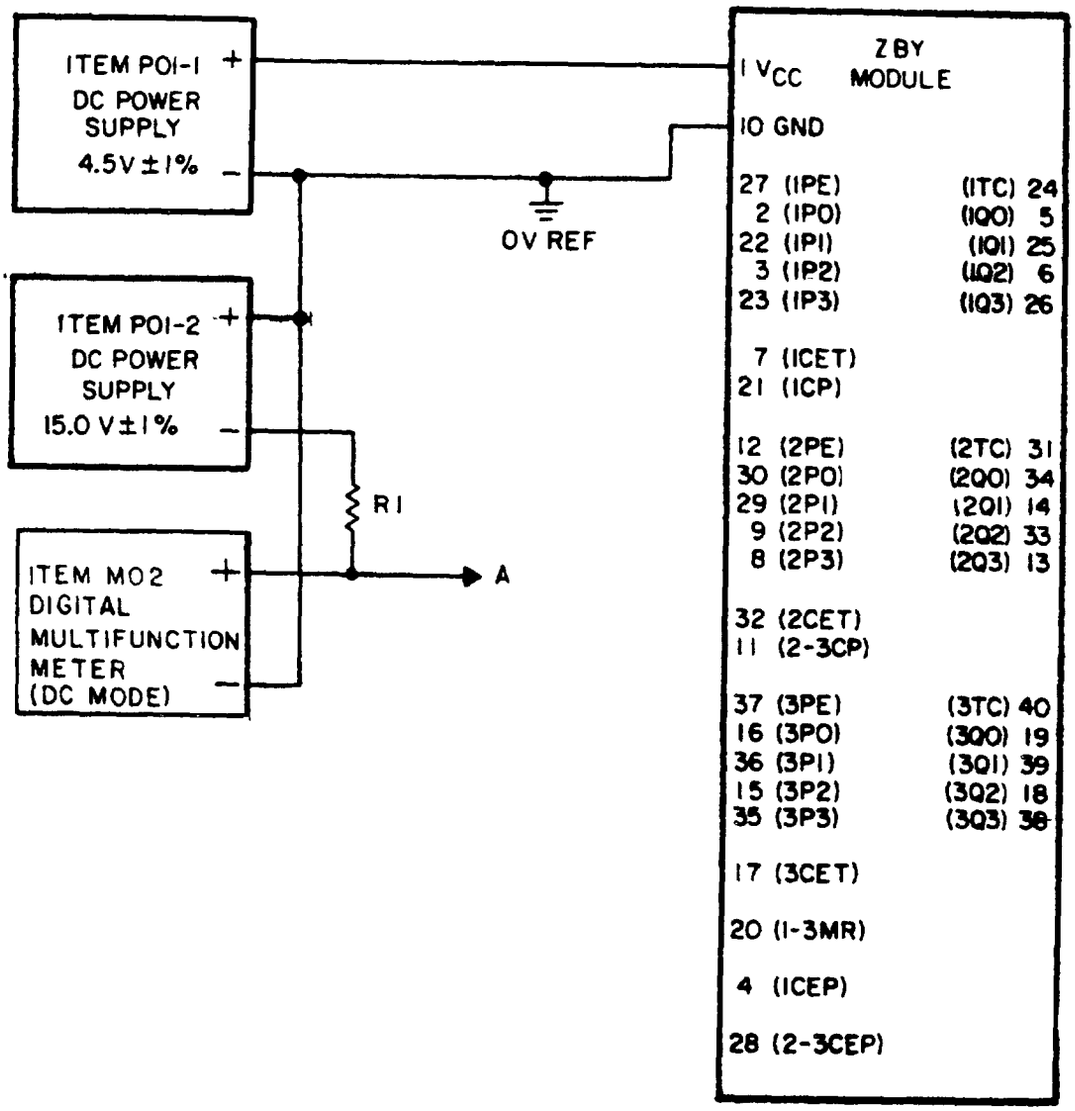
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
C1	1-3MR Input clamp diode voltage	20	-0.4	-1.5	V	7
C2	1CP	21				
C3	1P0	2				
C4	1P1	22				
C5	1P2	3				
C6	1P3	23				
C7	1CET	7				
C8	1PE	27				
C9	1CEP	4				
C10	2-3CP	11				
C11	2P0	30				
C12	2P1	29				
C13	2P2	9				
C14	2CET	32				
C15	2PE	12				
C16	2P3	8				
C17	3CET	17				
C18	3PE	37				
C19	3P3	35				

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TABLE VI. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C20	3P2 Input clamp diode voltage	15	-0.4	-1.5	V	7
C21	3P1	36	↓	↓	↓	↓
C22	3P0	16				
C23	2-3CEP	28				

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NOTE: R1 = 1.13 kΩ ±1%, 1/8 W.

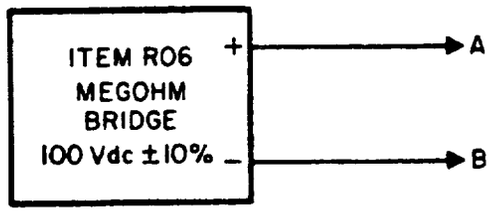
* FIGURE 7. Input clamp diode voltage test circuit for table VI.

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TABLE VII. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D	Isolation	All pins	Frame	10	-	MΩ	8

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1 V_{CC}	ZBY MODULE	
10 GND		
27 (1PE)	(1TC)	24
2 (1P0)	(1Q0)	5
22 (1P1)	(1Q1)	25
3 (1P2)	(1Q2)	6
23 (1P3)	(1Q3)	26
7 (1CET)		
21 (1CP)		
12 (2PE)	(2TC)	31
30 (2P0)	(2Q0)	34
29 (2P1)	(2Q1)	14
9 (2P2)	(2Q2)	33
8 (2P3)	(2Q3)	13
32 (2CET)		
11 (2-3CP)		
37 (3PE)	(3TC)	40
16 (3P0)	(3Q0)	19
36 (3P1)	(3Q1)	39
15 (3P2)	(3Q2)	18
35 (3P3)	(3Q3)	38
17 (3CET)		
20 (1-3MR)		
4 (1CEP)		
28 (2-3CEP)		

WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 8. Isolation test circuit for table VII.

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* TABLE VIII. Functional test.

Test cond	Parameter	Connect								Initial limits 25°±5°C ambient		Output logic states					Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max	TC	Q3	Q2	Q1	Q0		
E1	Functional	Open	G,2,3,22,23	4,7,27	Open	Open	21	B	20	See note 5, figure 9		See table XI						9
E2	Functional	Open	G,8,9,29,30	12,28,32	Open		11					See table XI						
E3	Functional	Open	G,15,16,35,36	17,28,37	Open		11					See table XI						
F1	Functional, low level	2,3,4,22,23,27	G	Open	7		21					L	L	L	L	L		
F2		2,3,4,22,23			7,27		21					L	H	H	H	H		
F3		8,9,12,28,29,30			32		11					L	L	L	L	L		
F4		8,9,28,29,30			12,32							L	H	H	H	H		
F5		15,16,28,35,36,37			17							L	L	L	L	L		
F6		15,16,28,35,36			17,37							L	H	H	H	H		
F7		2,3,7,22,23,27			4		21					L	L	L	L	L		
F8		8,9,12,29,30,32			28		11					L	L	L	L	L		
F9		15,16,17,35,36,37			28		11					L	L	L	L	L		

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* TABLE IX. Output voltage (high-level and low-level) test.

Test cond	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
G1	1TC Output voltage, high level	4,7	G,27	2,3,22,23	Open	24	21	B	20	2.4	4.5	V	9
G2	1Q3	Open	G,4,7,27	23	2,3,22	26							
G3	1Q2			3	2,22,23	6							
G4	1Q1			22	2,3,23	25							
G5	1Q0			2	3,22,23	5							
G6	2TC	28,32	G,12	8,9,29,30	Open	31	11						
G7	2Q3	Open	G,12,28,32	8	9,29,30	13							
G8	2Q2			9	8,29,30	33							
G9	2Q1			29	8,9,30	14							
G10	2Q0			30	8,9,29	34							
G11	3TC	28,17	G,37	15,16,35,36	Open	40							
G12	3Q3	Open	G,17,28,37	35	15,16,36	38							
G13	3Q2			15	16,35,36	18							
G14	3Q1			36	15,16,35	39							
G15	3Q0			16	15,35,36	19							
H1	1Q3 Output voltage, low level		G,4,7,27	2,3,22	23	26	21			0	300	mV	
H2	1Q2			2,22,23	3	6							
H3	1Q1			2,3,23	22	25							
H4	1Q0			3,22,23	2	5							

* TABLE IX. Output voltage (high-level and low-level) test - Continued.

Test cond	Parameter	Connect								Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max		
H5	2Q3 Output voltage, low level	Open	G,12,28, 32	9,29,30	8	13	11	8	20	0	300	mV	9
H6	2Q2			8,29,30	9	33							
H7	2Q1			8,9,30	29	14							
H8	2Q0			8,9,29	30	34							
H9	3Q3		G,17,28, 37	15,16,36	35	38							
H10	3Q2			16,35,36	15	18							
H11	3Q1			15,16,35	36	39							
H12	3Q0			15,35,36	16	19							
H13	1TC		6,4,7,27	2,3,23	22	24	21						
H14	2TC		6,12,28, 32	8,9,30	29	31	11						
H15	3TC		6,17,28, 37	15,35,36	16	40	11						

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*TABLE X. Input threshold (low-level and high-level) check test.

Test cond	Parameter	Connect								Initial limits 25°±5°C ambient		Output logic states					Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	G to pin	H to pin	Min	Max	TC	Q3	Q2	Q1	Q0	
J1	1-3MR Input threshold low level check	2,3,4,7,22,23	27	Open	G	Open	20	D	21	See note 5, figure 9		L	L	L	L	L	9
J2	1-3MR Input threshold low level check	8,9,28,29,30, 32	12	Open	G		20	D	11			L	L	L	L	L	
J3	1-3MR Input threshold low level check	15,16,17,28, 35,36	37	Open	G		20	D	11			L	L	L	L	L	
K1	1CET Input threshold high level check	4,27	G,2,3,22,23	7	Open		21	B	20			L	L	L	L	H	
K2	2CET	12,28	G,8,9,29,30	32			11					L	L	L	L	H	
K3	3CET	28,37	G,15,16,35,36	17			11					L	L	L	L	H	
K4	1CEP	7,27	G,2,3,22,23	4			21					L	L	L	L	H	
K5	2-3CEP	12,32	G,8,9,29,30	28			11					L	L	L	L	H	
K6	2-3CEP	17,37	G,15,35,36	28			11					L	L	L	L	H	

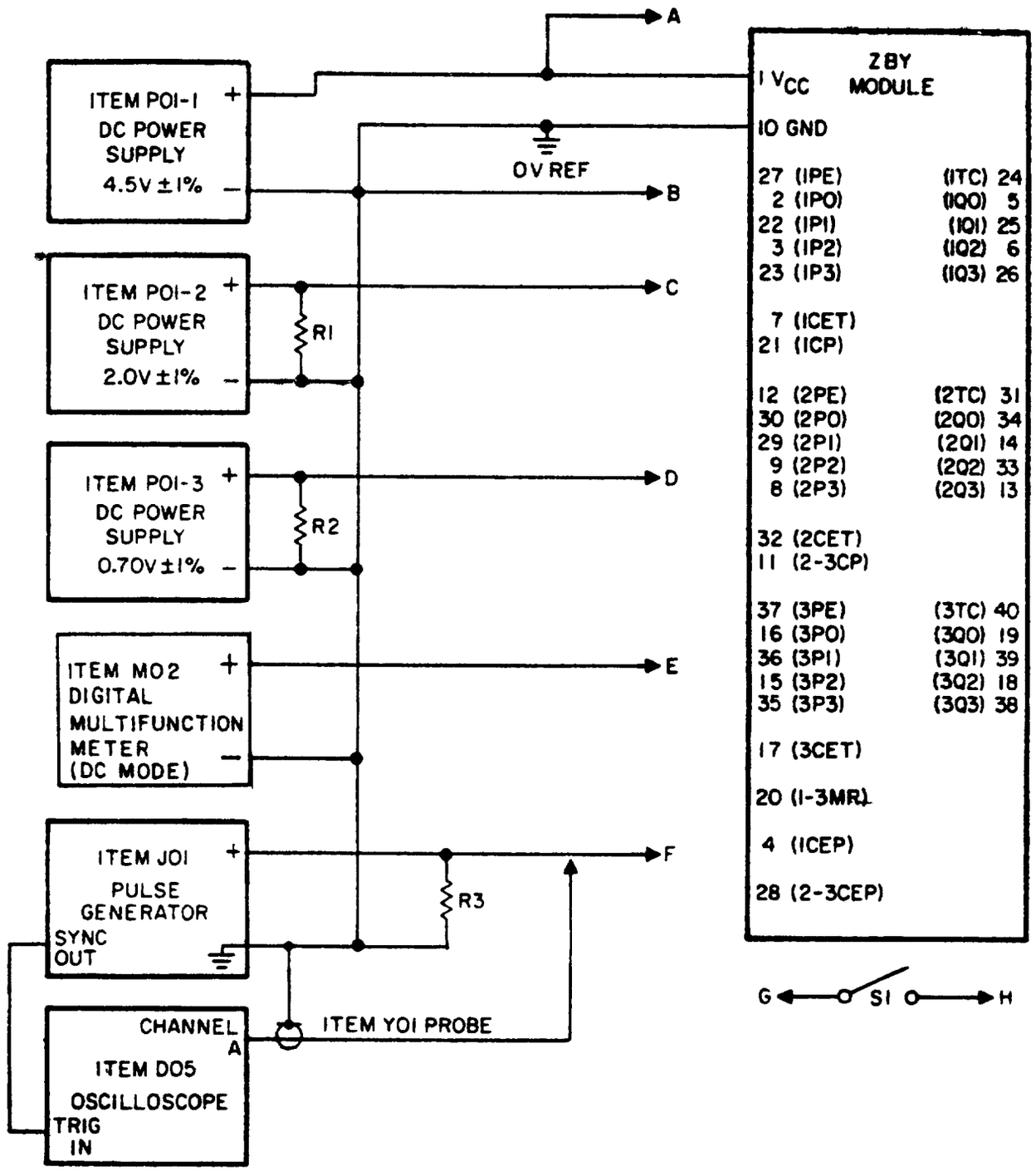
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TABLE XI. Output logic states for test conditions E1-E3. 1/, 2/

Test number	Total pulses	Q0	Q1	Q2	Q3	TC
Initial state	0	L	L	L	L	L
1	1	H	L	L	L	L
2	2	L	H	L	L	L
3	3	H	H	L	L	L
4	4	L	L	H	L	L
5	5	H	L	H	L	L
6	6	L	H	H	L	L
7	7	H	H	H	L	L
8	8	L	L	L	H	L
9	9	H	L	L	H	L
10	10	L	H	L	H	L
11	11	H	H	L	H	L
12	12	L	L	H	H	L
13	13	H	L	H	H	L
14	14	L	H	H	H	L
15	15	H	H	H	H	H
16	16	L	L	L	L	L

1/ H = High level.
2/ L = Low level.

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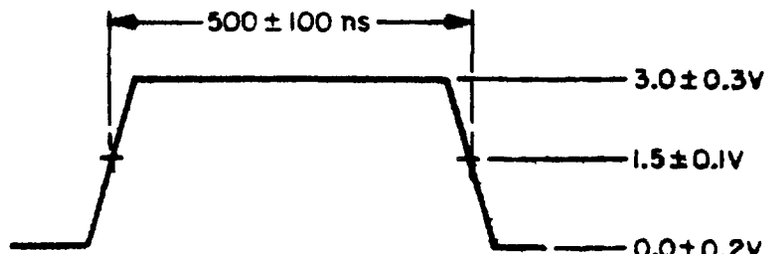


* FIGURE 9. Functional, output voltage (high-level and low-level), and input threshold (low-level and high-level) check test circuit for tables VIII, IX and X.

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NOTES:
1. R1-R3 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Adjust the input pulse width, transition times, and pulse amplitude using item D05 with an item J01 pulse repetition rate of 30 kHz ±50%. Adjust the dc logic level with item J01 in the single pulse mode using item M02.
4. For test conditions G1-G15 and H1-H15 attach an output logic load circuit to each of the following pins: 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40. Each output logic load shall be as specified on figure 10, load circuit 1.
5. Low level = 0.0 to 0.3 V.
   High level = 2.4 to 4.5 V.
   For test conditions E, F, J, and K, verification of the above logic levels is obtained by connecting item M02 to each output pin in succession after input conditions are accomplished.
6. Test conditions E1-E3:

   a. Input pulse at point F with F open.



$t_{TLH} = t_{THL} = 50 \pm 25$ ns

   b. After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With item J01 in the single pulse mode, apply 16 pulses to the circuit under test. The output logic states for each input pulse shall be as specified in table XI.

7. Test conditions F1-F9:

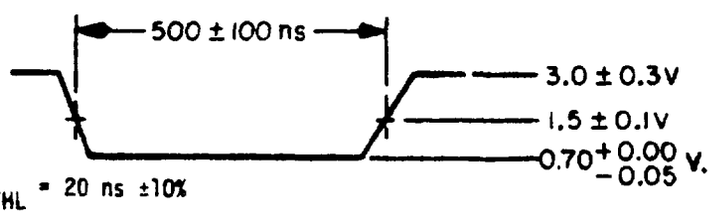
   a. The input pulse at point F with F open shall be as specified in note 6.
   b. After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table VIII.

8 Test conditions G1-G15 and H1-H15·

   a. The input pulse at point F with F open shall be as specified in note 6.
   b. After making all circuit connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The voltage levels shall be as specified in table IX

9. Test conditions J1-J3:

   a. Input pulse at point F with F open.



$t_{TLH} = t_{THL} = 20$ ns ±10%

* FIGURE 9. Functional, output voltage (high-level and low-level), and input threshold (low-level and high-level) check test circuit for tables VIII, IX and X - Continued.

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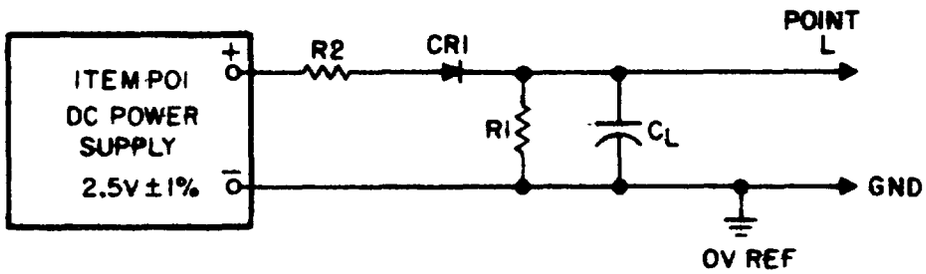
b. Initially switch S1 shall be closed. After making all circuit connections for a test condition, open switch S1 for a minimum of 100 ns, then close switch S1. With item J01 in the single pulse mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table VIII.

10. Test conditions K1-K6:

a. The input pulse at point F with F open shall be as specified in note 6.
b. After making all connections for a test condition, close switch S1 for a minimum of 100 ns, then open switch S1. With item J01 in the single mode, apply one pulse to the circuit under test. The output logic states shall be as specified in table X.

* FIGURE 9. Functional, output voltage (high-level and low-level), and input threshold (low-level and high-level) check test circuit for tables VIII, IX and X - Continued.

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NOTES:

1. Load circuit 1 for test conditions G1-G15, H1-H15, S1-S15, U1-U15, W1-W15, Y1-Y15, AA1-AA15, CC1-CC6, EE1-EE6, FF1-FF6, GG1-GG27, HH1-HH15, and JJ1-JJ30:

   a. CR1 = 1N4150, or equivalent.
   b. C_L = 40 ±5 pF (including probe and parasitics capacitance).
   c. R1 = Select value for a current flow of 0.32 mA ±1% into point L when point L is held at 2.4 V (7.5 kΩ).
   d. R2 = Select value for a current flow of 3.2 mA ±1% when point L is held at 0.3 V.
   e. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions HH1-HH15 and JJ1-JJ30.

2. Load circuit 2 for test conditions T1-T15, V1-V15, X1-X15, Z1-Z15, BB1-BB15, and DD1-DD6:

   a. Same as 1.a.
   b. C_L = 100 ±5 pF (including probe and parasitics capacitance).
   c. Same as 1 c.
   d. Same as 1 d.

FIGURE 10. ZBY module logic load circuit.

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TABLE XII. Input current (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
L1	1-3MR Input current, low level	Open	Open	20	Open	0.82	1.88	mA	11
L2	1CP			21		0.10	0.86		
L3	1CET			7					
L4	1PE			27					
L5	2-3CEP			28					
L6	2CET			32					
L7	3CET			17					
L8	2PE			12					
L9	3PE			37					
L10	2-3CP			11			1.72		
L11	1CEP			4			0.43		
L12	1P0			2	27		0.29		
L13	1P1			22					
L14	1P2			3					
L15	1P3			23					
L16	2P0			30	12				
L17	2P1			29					
L18	2P2			9					
L19	2P3			8					
L20	3P3			35	37				
L21	3P2			15					
L22	3P1			36					
L23	3P0			16					

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TABLE XII. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
M1	1-3MR Input current, high level	20	Open	Open	2,3,4,7,8,9,11,12,15,16,17, 21,22,23,27,28,29,30,32,35, 36,37	-228	-336	µA	11
M2	1CP	21			2,3,4,7,20,22,23,27	0	40		
M3	1CET	7			2,3,4,20,21,22,23,27				
M4	1PE	27			2,3,4,7,20,21,22,23				
M5	2-3CEP	28			8,9,11,12,15,16,17,20,29,30, 32,35,36,37				
M6	2CET	32			8,9,11,12,20,28,29,30				
M7	3CET	17			11,15,16,20,28,35,36,37				
M8	2PE	12			8,9,11,20,28,29,30,32				
M9	3PE	37			11,15,16,17,20,28,35,36				
M10	2-3CP	11			8,9,12,15,16,17,20,28,29,30, 32,35,36,37		80		
M11	1CEP	4			2,3,7,20,21,22,23,27		20		
M12	1P0	2			3,4,7,20,21,22,23,27	0.0	13.3		
M13	1P1	22			2,3,4,7,20,21,23,27				
M14	1P2	3			2,4,7,20,21,22,23,27				
M15	1P3	23			2,3,4,7,20,21,22,27				
M16	2P0	30			8,9,11,12,20,28,29,32				
M17	2P1	29			8,9,11,12,20,28,30,32				
M18	2P2	9			8,11,12,20,28,29,30,32				
M19	2P3	8			9,11,12,20,28,29,30,32				
M20	3P3	35			11,15,16,17,20,28,36,37				

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TABLE XII. Input current (low-level and high-level) test - Continued.

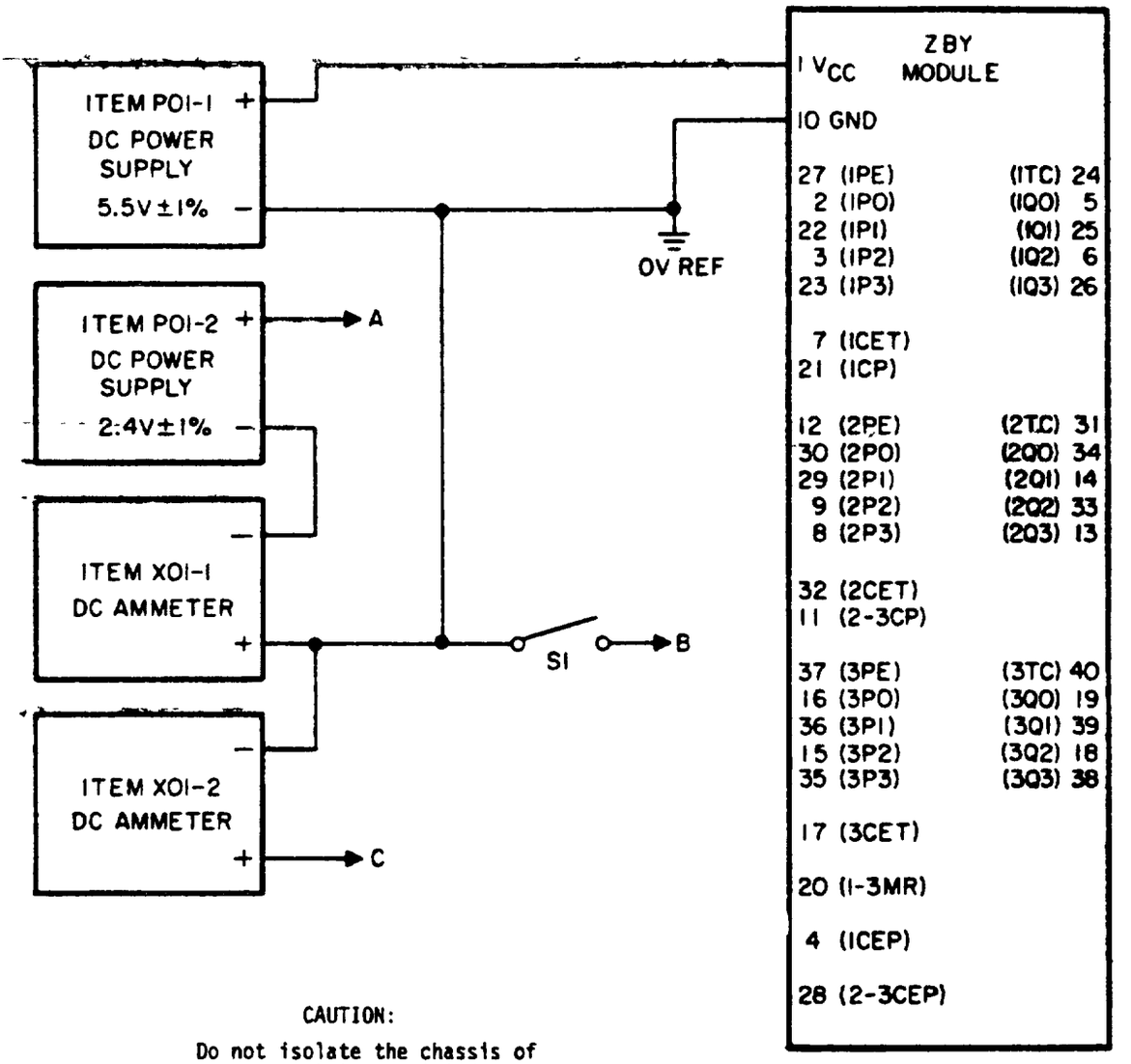
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
M21	3P2 Input current, high level	15	Open	Open	11,16,17,20,28,35,36,37	0.0	13.3	µA	11
M22	3P1 Input current, high level	36	Open	Open	11,15,16,17,20,28,35,37	0.0	13.3	µA	11
M23	3P0 Input current, high level	16	Open	Open	11,15,17,20,28,35,36,37	0.0	13.3	µA	11

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TABLE XIII. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin 10 to pin	Min	Max		
N1	1Q3 Output short circuit current	Open	11,12	26	12,27,37	2.5	25.0	mA	11
N2	1Q2			6					
N3	1Q1			25					
N4	1Q0			5					
N5	1TC			24					
N6	2Q3			13					
N7	2Q2			33					
N8	2Q1			14					
N9	2Q0			34					
N10	2TC			31					
N11	3Q2			18					
N12	3Q3			38					
N13	3Q1			39					
N14	3Q0			19					
N15	3TC			40					

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CAUTION:

Do not isolate the chassis of ammeter X01-1 or X01-2 from earth ground.

NOTES:
1. S1 = SPST switch.
2. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
3. After making all circuit connections for a test condition except point C, close switch S1 for a minimum of 100 ns, then open switch S1. Then connect point C and take measurement.
4. Only one output shall be shorted at a time, and the measurement taken within 10 seconds

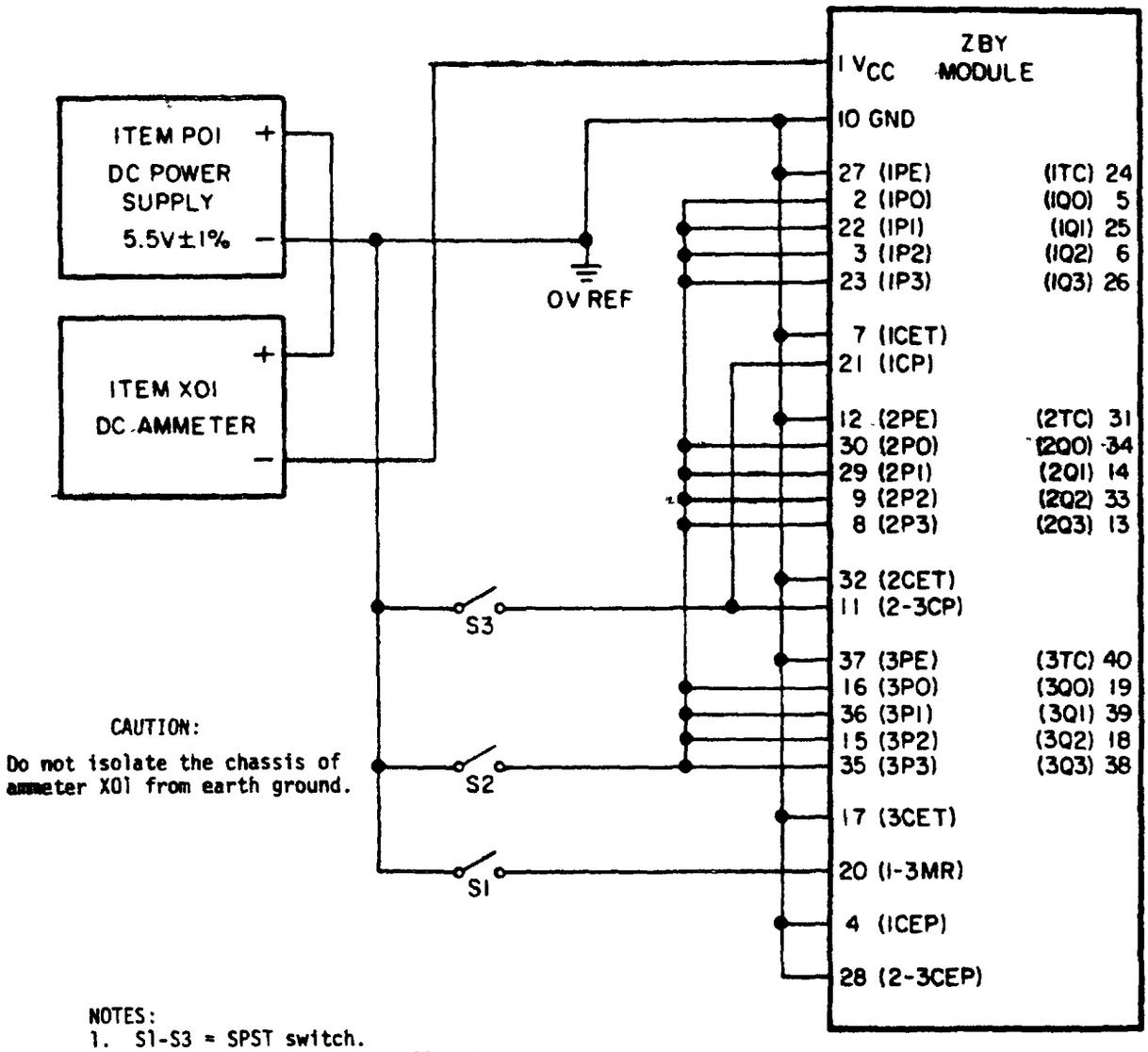
FIGURE 11. Input current (low-level and high-level) and output short circuit current test circuit for tables XII and XIII.

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* TABLE XIV. Power supply current (high-level and low-level) test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
P1	Power supply current, high level	1.0	82.5	mA	12
P2	Power supply current, low level	1.0	82.5	mA	12

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NOTES:
1. S1-S3 = SPST switch.
2. Power supply current - all outputs high:

   a. Switch S1 open.
   b. Initially switch S2 shall be open and switch S3 closed.
   c. Open switch S3 for 100 ns minimum, then close switches S3 and S2.
   d. Take measurement.

3. Power supply current - all outputs low:

   a. Close switches S1, S2, and S3.
   b. Take measurement.

* FIGURE 12. Power supply current (high-level and low-level) test circuit for table XIV.

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TABLE XV. Input capacitance test.

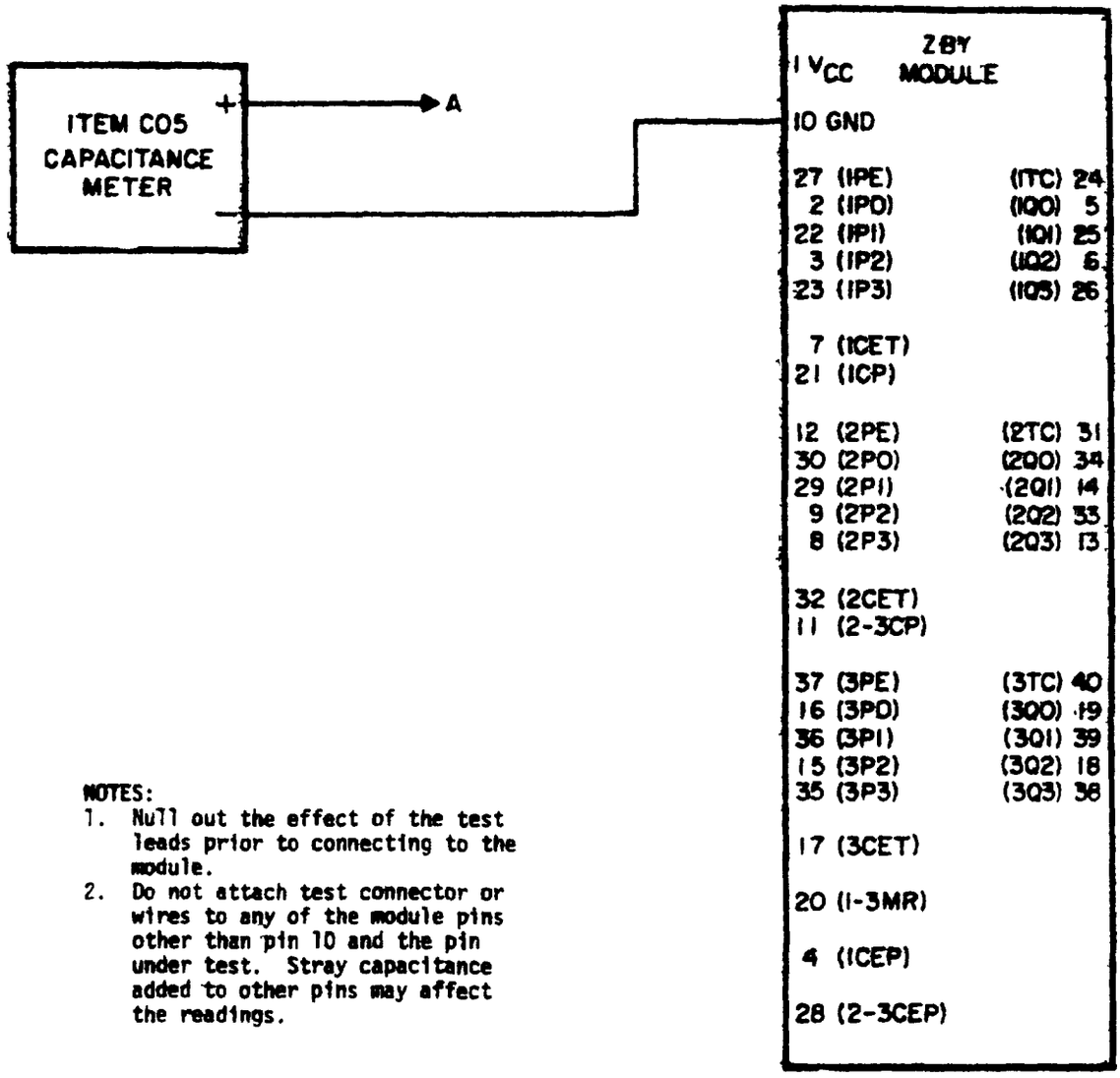
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q1	1-3MR Input capacitance	20	3	45	pF	13
Q2	2-3CEP	28	2	30		
Q3	2-3CP	11		30		
Q4	1PE	27		15		
Q5	1CP	21				
Q6	1PO	2				
Q7	1P1	22				
Q8	1P2	3				
Q9	1P3	23				
Q10	1CET	7				
Q11	1CEP	4				
Q12	2PE	12				
Q13	2PO	30				
Q14	2P1	29				
Q15	2P2	9				
Q16	2P3	8				
Q17	2CET	32				
Q18	3PO	16				
Q19	3P1	36				

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TABLE XV. Input capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
Q20	3P2	Input capacitance	15	2	15	pF	13
Q21	3P3	↓	35	↓	↓	↓	↓
Q22	3CET	↓	17	↓	↓	↓	↓
Q23	3PE	↓	37	↓	↓	↓	↓

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

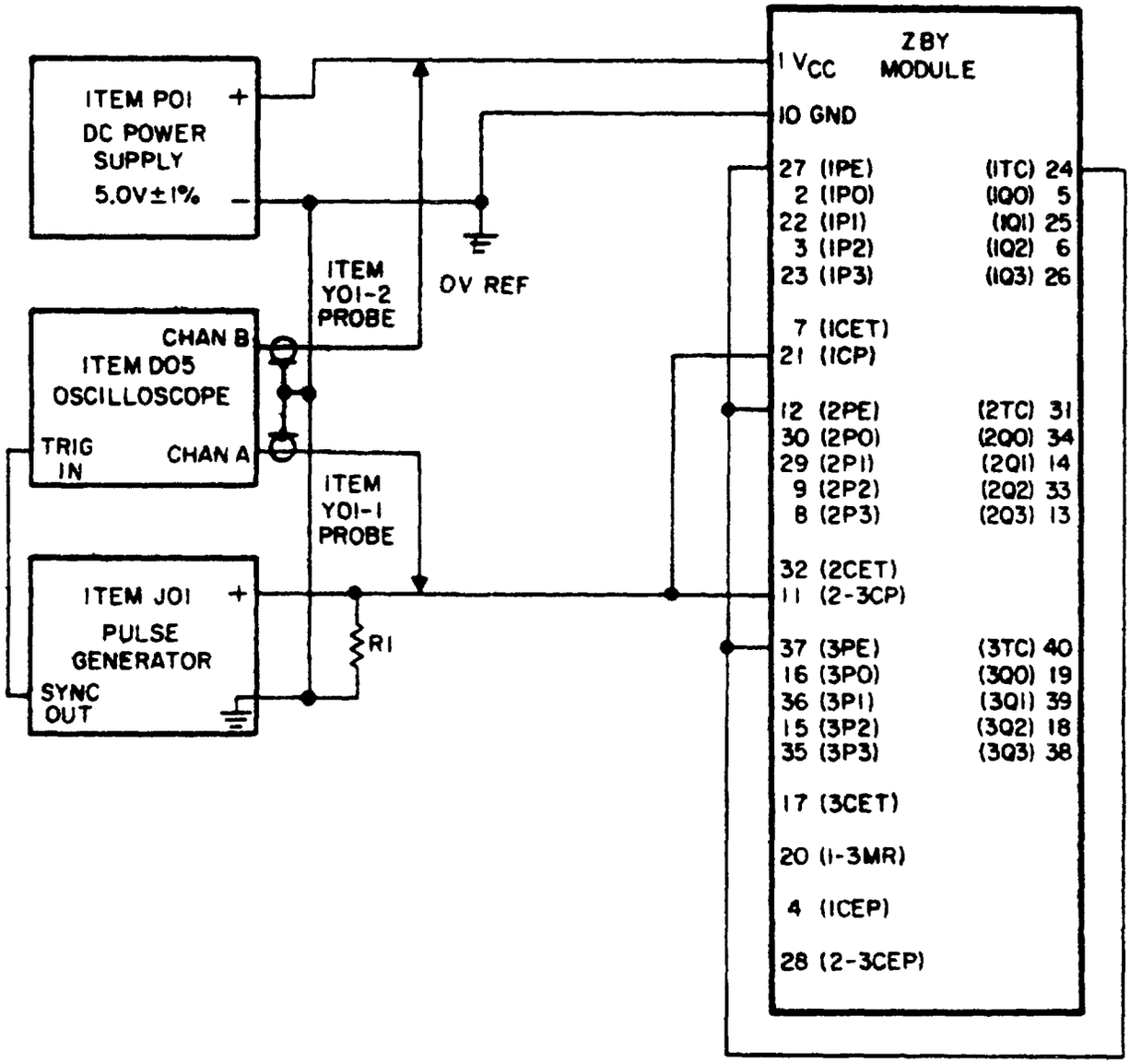
FIGURE 13. Input capacitance test circuit for table XV

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
R	Power supply transient amplitude	0	400	mV	14,15

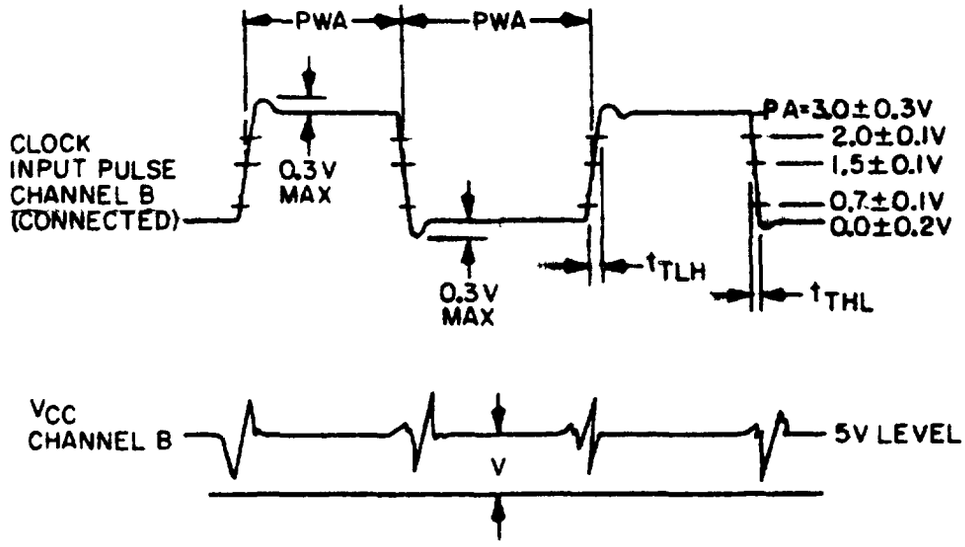
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NOTE: R1 = 51Ω ±5%, 1/2 W.

* FIGURE 14. Power supply transient amplitude test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
2. The value V is measured from the baseline to the most negative excursion of the waveform, which may occur in time coincident with either of two pulses.

* FIGURE 15. Power supply transient amplitude test circuit waveforms for table XVI.

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* TABLE XVII. Propagation delay time (t_{PHL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: Pin to pin		Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
S1	1CP to 1TC Propagation delay time (t_{PHL}), C_L = 40 pF	Open	21	24	25	27	10	49	ns	16,17
S2	2-3CEP to 2TC		11	31	14	12	10	49		
S3	2-3CP to 3TC		11	40	39	37	10	49		
S4	1CP to 1Q0		21	5	25	27	15	59		
S5	1CP to 1Q3			26	25					
S6	1CP to 1Q1			25	5					
S7	1CP to 1Q2			6	25					
S8	2-3CP to 2Q0		11	34	14	12				
S9	2-3CP to 2Q3			13	14					
S10	2-3CP to 2Q1			14	34					
S11	2-3CP to 2Q2			33	14					
S12	2-3CP to 3Q0			19	39	37				
S13	2-3CP to 3Q3			38	39					
S14	2-3CP to 3Q1			39	19					
S15	2-3CP to 3Q2			18	39					
T1	1CP to 1TC Propagation delay time (t_{PHL}), C_L = 100 pF		21	24	25	27	10	55		
T2	2-3CP to 2TC		11	31	14	12	10	55		
T3	2-3CP to 3TC		11	40	39	37	10	55		
T4	1CP to 1Q0		21	5	25	27	15	65		
T5	1CP to 1Q3		21	26	25					
T6	1CP to 1Q1		21	25	5					

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* TABLE XVII. Propagation delay time (t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin		Min	Max		
T7	1CP to 1Q2 Propagation delay time (t_{PHL}), C_L = 100 pF	Open	21	6	25	27	15	65	ns	16,17
T8	2-3CP to 2Q0		11	34	14	12				
T9	2-3CP to 2Q3			13	14					
T10	2-3CP to 2Q1			14	34					
T11	2-3CP to 2Q2			33	14					
T12	2-3CP to 3Q0			19	39	37				
T13	2-3CP to 3Q3			38	39					
T14	2-3CP to 3Q1			39	19					
T15	2-3CP to 3Q2			18	39					

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* TABLE XVIII. Transition time (t_{THL}) test.

Test condition		Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin to pin		Min	Max		
U1	1TC	Transition time (t_{THL}), C_L = 40 pF	Open	21	24	25	27	2	20	ns	16,17
U2	2TC			11	31	14	12				
U3	3TC			11	40	39	37				
U4	1Q0			21	5	25	27				
U5	1Q3				26	25					
U6	1Q1				25	5					
U7	1Q2				6	25					
U8	2Q0			11	34	14	12				
U9	2Q3				13	14					
U10	2Q1				14	34					
U11	2Q2				33	14					
U12	3Q0				19	39	37				
U13	3Q3				38	39					
U14	3Q1				39	19					
U15	3Q2				18	39					
V1	1TC	Transition time (t_{THL}), C_L = 100 pF		21	24	25	27		25		
V2	2TC			11	31	14	12				
V3	3TC			11	40	39	37				
V4	1Q0			21	5	25	27				
V5	1Q3				26	25	27				
V6	1Q1				25	5	27				

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* TABLE XVIII. Transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin to pin		Min	Max		
V7	1Q2	Transition time (t_{THL}), C_L = 100 pF	Open	21	6	25	27	2	25	ns	16,17
V8	2Q0	↓	↓	11	34	14	12	↓	↓	↓	↓
V9	2Q3	↓	↓	↓	13	14	↓	↓	↓	↓	↓
V10	2Q1	↓	↓	↓	14	34	↓	↓	↓	↓	↓
V11	2Q2	↓	↓	↓	33	14	↓	↓	↓	↓	↓
V12	3Q0	↓	↓	↓	19	39	37	↓	↓	↓	↓
V13	3Q3	↓	↓	↓	38	39	↓	↓	↓	↓	↓
V14	3Q1	↓	↓	↓	39	19	↓	↓	↓	↓	↓
V15	3Q2	↓	↓	↓	18	39	↓	↓	↓	↓	↓

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* TABLE XIX. Propagation delay time (t_{PLH}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: Pin to pin		Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
W1	1CP to 1TC Propagation delay time (t_{PLH}), C_L = 40 pF	2	21	24	25	27	20	99	ns	16,18
W2	2-3CP to 2TC	30	11	31	14	12	20	99		
W3	2-3CP to 3TC	16	11	40	39	37	20	99		
W4	1CP to 1Q0	2	21	5	26	27	10	54		
W5	1CP to 1Q3	23		26	25					16,17
W6	1CP to 1Q1	22		25	5					16,17
W7	1CP to 1Q2	3		6	25					16,17
W8	2-3CP to 2Q0	30	11	34	13	12				16,18
W9	2-3CP to 2Q3	8		13	14					16,17
W10	2-3CP to 2Q1	29		14	34					16,17
W11	2-3CP to 2Q2	9		33	14					16,17
W12	2-3CP to 3Q0	16		19	38	37				16,18
W13	2-3CP to 3Q3	35		38	39					16,17
W14	2-3CP to 3Q1	36		39	19					16,17
W15	2-3CP to 3Q2	15		18	39					16,17
X1	1CP to 1TC Propagation delay time (t_{PLH}), C_L = 100 pF	2	21	24	25	27	25	105		16,18
X2	2-3CP to 2TC	30	11	31	14	12	25	105		
X3	2-3CP to 3TC	16	11	40	39	37	25	105		
X4	1CP to 1Q0	2	21	5	26	27	15	60		
X5	1CP to 1Q3	23	21	26	25	27	15	60		16,17
X6	1CP to 1Q1	22	21	25	5	27	15	60		16,17

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* TABLE XIX. Propagation delay time (t_{PLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin		Min	Max		
X7	1CP to 1Q2 Propagation delay time (t_{PLH}), C_L = 100 pF	3	21	6	25	27	15	60	ns	16,17
X8	2-3CP to 2Q0	30	11	34	13	12				16,18
X9	2-3CP to 2Q3	8		13	14					16,17
X10	2-3CP to 2Q1	29		14	34					16,17
X11	2-3CP to 2Q2	9		33	14					16,17
X12	2-3CP to 3Q0	16		19	38	37				16,18
X13	2-3CP to 3Q3	35		38	39					16,17
X14	2-3CP to 3Q1	36		39	19					16,17
X15	2-3CP to 3Q2	15		18	39					16,17

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* TABLE XX. Transition time (t_{TLH}) test.

Test condition	Parameter		Connect					Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Pin to pin		Min	Max		
Y1	1TC	Transition time (t_{TLH}), C_L = 40 pF	2	21	24	25	27	5	45	ns	16,18
Y2	2TC		30	11	31	14	12				
Y3	3TC		16	11	40	39	37				
Y4	1Q0		2	21	5	26	27				
Y5	1Q3		23		26	25					16,17
Y6	1Q1		22		25	5					16,17
Y7	1Q2		3		6	25					16,17
Y8	2Q0		30	11	34	13	12				16,18
Y9	2Q3		8		13	14					16,17
Y10	2Q1		29		14	34					16,17
Y11	2Q2		9		33	14					16,17
Y12	3Q0		16		19	38	37				16,18
Y13	3Q3		35		38	39					16,17
Y14	3Q1		36		39	19					16,17
Y15	3Q2		15		18	39					16,17
Z1	1TC	Transition time (t_{TLH}), C_L = 100 pF	2	21	24	25	27	10	55		16,18
Z2	2TC		30	11	31	14	12				
Z3	3TC		16	11	40	39	37				
Z4	1Q0		2	21	5	26	27				
Z5	1Q3		23		26	25					16,17
Z6	1Q1		22		25	5					16,17
Z7	1Q2		3		6	25					16,17

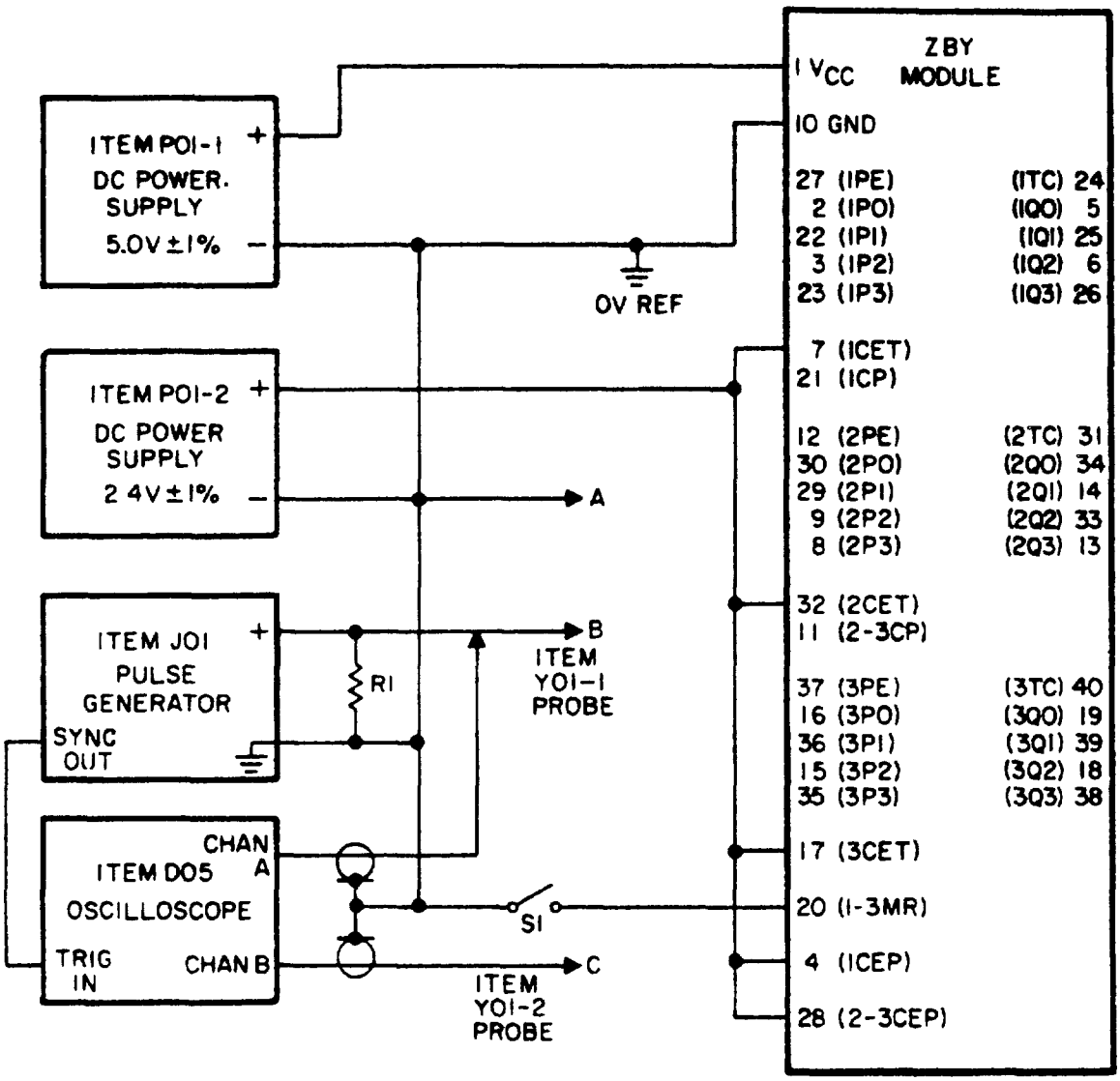
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* TABLE XX. Transition time (t_{TLH}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Pin to pin		Min	Max		
Z8	2Q0 Transition time (t_{TLH}), C_L = 100 pF	30	11	34	13	12	10	55	ns	16,18
Z9	2Q3	8	↓	13	14	↓	↓	↓	↓	16,17
Z10	2Q1	29	↓	14	34	↓	↓	↓	↓	16,17
Z11	2Q2	9	↓	33	14	↓	↓	↓	↓	16,17
Z12	3Q0	16	↓	19	38	37	↓	↓	↓	16,18
Z13	3Q3	35	↓	38	39	↓	↓	↓	↓	16,17
Z14	3Q1	36	↓	39	19	↓	↓	↓	↓	16,17
Z15	3Q2	15	↓	18	39	↓	↓	↓	↓	16,17

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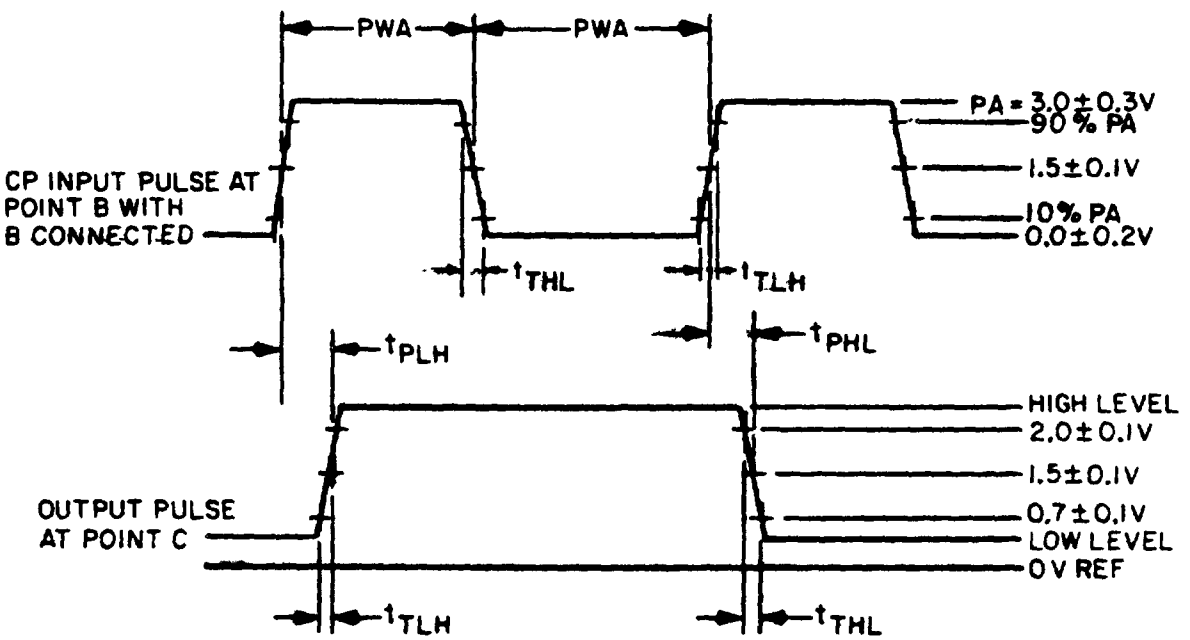
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NOTES
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. For test conditions S, U, W, and Y load all outputs as specified on figure 10, load circuit 1.
4. For test conditions T, V, X, and Z load all outputs as specified on figure 10, load circuit 2
5 After making all circuit connections for a test condition, momentarily close switch S1 for a minimum of 100 ns.

FIGURE 16. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{THL} and t_{TLH}) test circuit for tables XVII through XX.

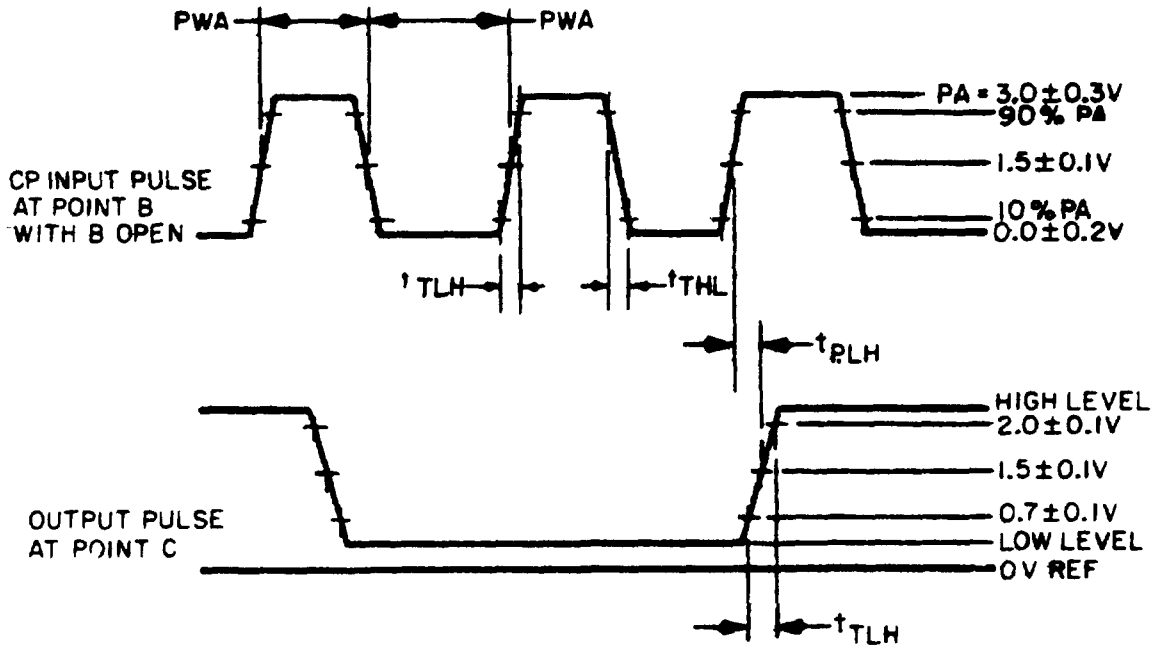
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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Before making the measurements specified, verify that the output pulse low level is 0.3 V maximum and the high level is 2.4 V minimum.
3. All voltage levels are referenced to ground.

* FIGURE 17. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{THL} and t_{TLH}) test circuit waveforms for tables XVII through XX.

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NOTES
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. Before making the measurements specified, verify that the output pulse low level is 0.3 V maximum and the high level is 2.4 V minimum.
3 All voltage levels are referenced to ground.

* FIGURE 18. Propagation delay time (t_{PHL} and t_{PLH}) and transition time (t_{THL} and t_{TLH}) test circuit waveforms for tables XIX and XX.

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*TABLE XXI. Propagation delay time (t_{PHL}) test and minimum pulse width and recovery time check test.

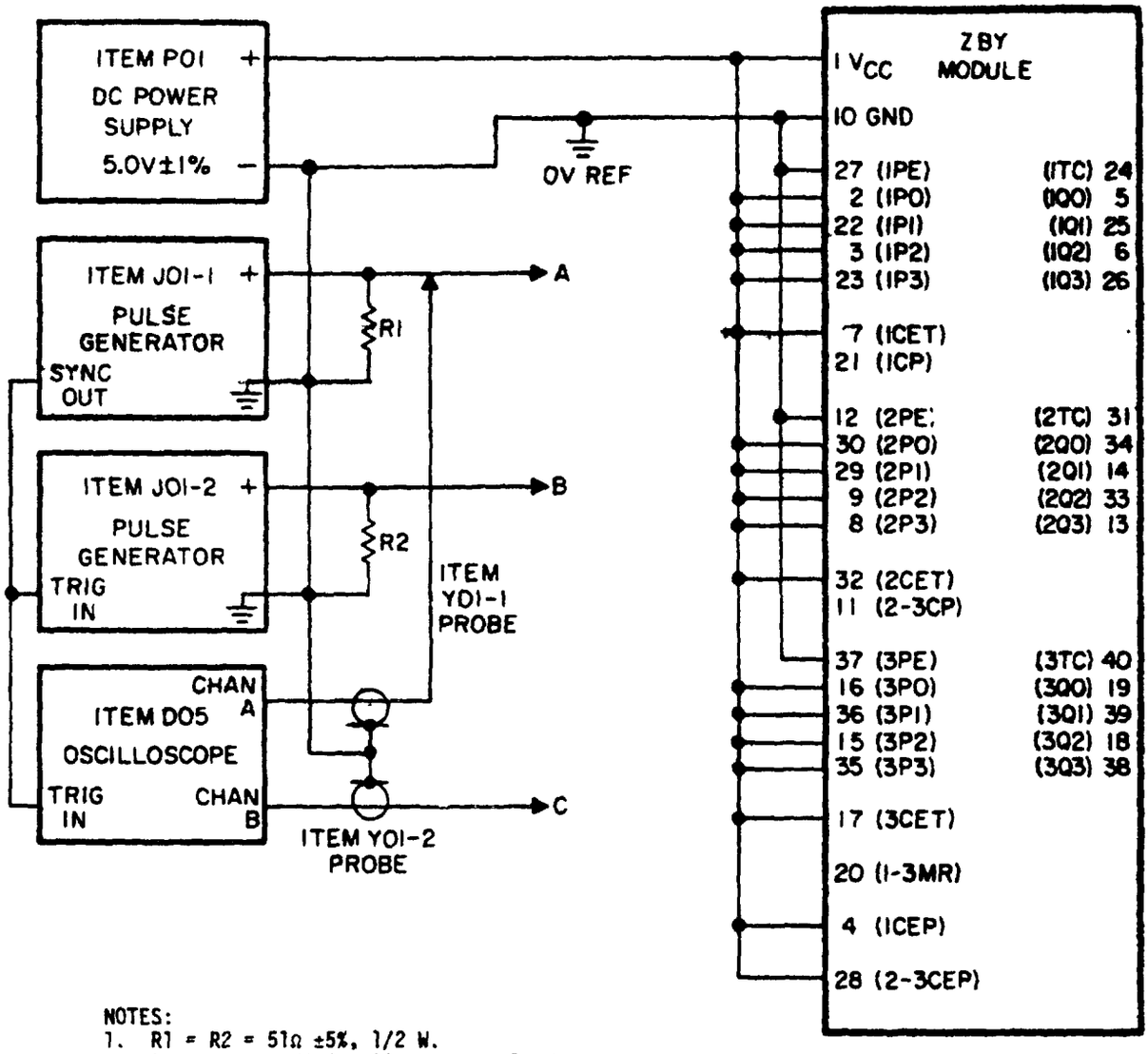
Test condition	Parameter		Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
AA1	1-3MR to 1Q3	Propagation delay time (t_{PHL}) and minimum pulse width and recovery time check, C_L = 40 pF	20	21	26	25	99	ns	19,20
AA2	1-3MR to 1Q2				6				
AA3	1-3MR to 1Q1				25				
AA4	1-3MR to 1Q0				5				
AA5	1-3MR to 1TC				24				
AA6	1-3MR to 2Q3			11	13				
AA7	1-3MR to 2Q2				33				
AA8	1-3MR to 2Q1				14				
AA9	1-3MR to 2Q0				34				
AA10	1-3MR to 2TC				31				
AA11	1-3MR to 3Q3				38				
AA12	1-3MR to 3Q2				18				
AA13	1-3MR to 3Q1				39				
AA14	1-3MR to 3Q0				19				
AA15	1-3MR to 3TC				40				
BB1	1-3MR to 1Q3	Propagation delay time (t_{PHL}) and minimum pulse width and recovery time check, C_L = 100 pF		21	26		105		
BB2	1-3MR to 1Q2				6				
BB3	1-3MR to 1Q1				25				
BB4	1-3MR to 1Q0				5				
BB5	1-3MR to 1TC				24				

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*TABLE XXI. Propagation delay time (t_{PHL}) test and minimum pulse width and recovery time check test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
BB6	1-3MR to 2Q3 Propagation delay time (t_{PHL}) and minimum pulse width and recovery time check, C_L = 100 pF	20	11	13	25	105	ns	19,20
BB7	1-3MR to 2Q2			33				
BB8	1-3MR to 2Q1			14				
BB9	1-3MR to 2Q0			34				
BB10	1-3MR to 2TC			31				
BB11	1-3MR to 3Q3			38				
BB12	1-3MR to 3Q2			18				
BB13	1-3MR to 3Q1			39				
BB14	1-3MR to 3Q0			19				
BB15	1-3MR to 3TC			40				

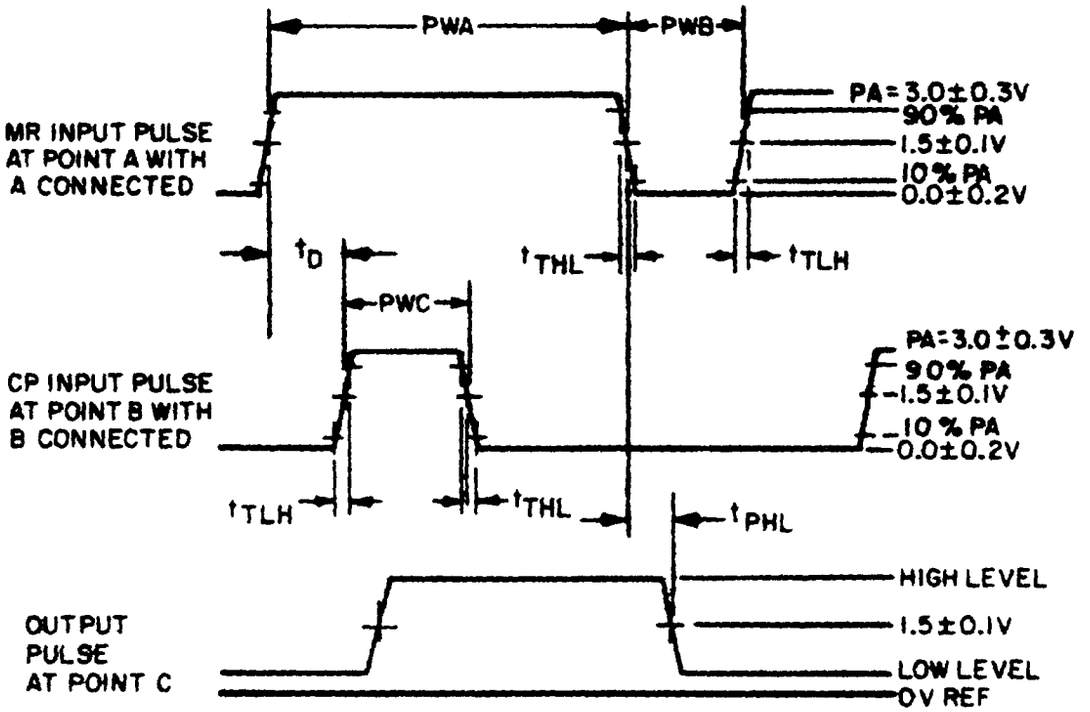
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. For test condition AA, connect loads to all outputs as specified on figure 10, load circuit 1.
3. For test condition BB, connect loads to all outputs specified on figure 10, load circuit 2.

* FIGURE 19. Propagation delay time (t_{PHL}) test and minimum pulse width and recovery time check test circuit for table XXI.

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NOTES:
1. Input pulse parameters:
   PWA = 300 ±30 ns.
   PWB = 65 ±2 ns.
   PWC = 100 ±10 ns.
   t_D = 75 ±5 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.

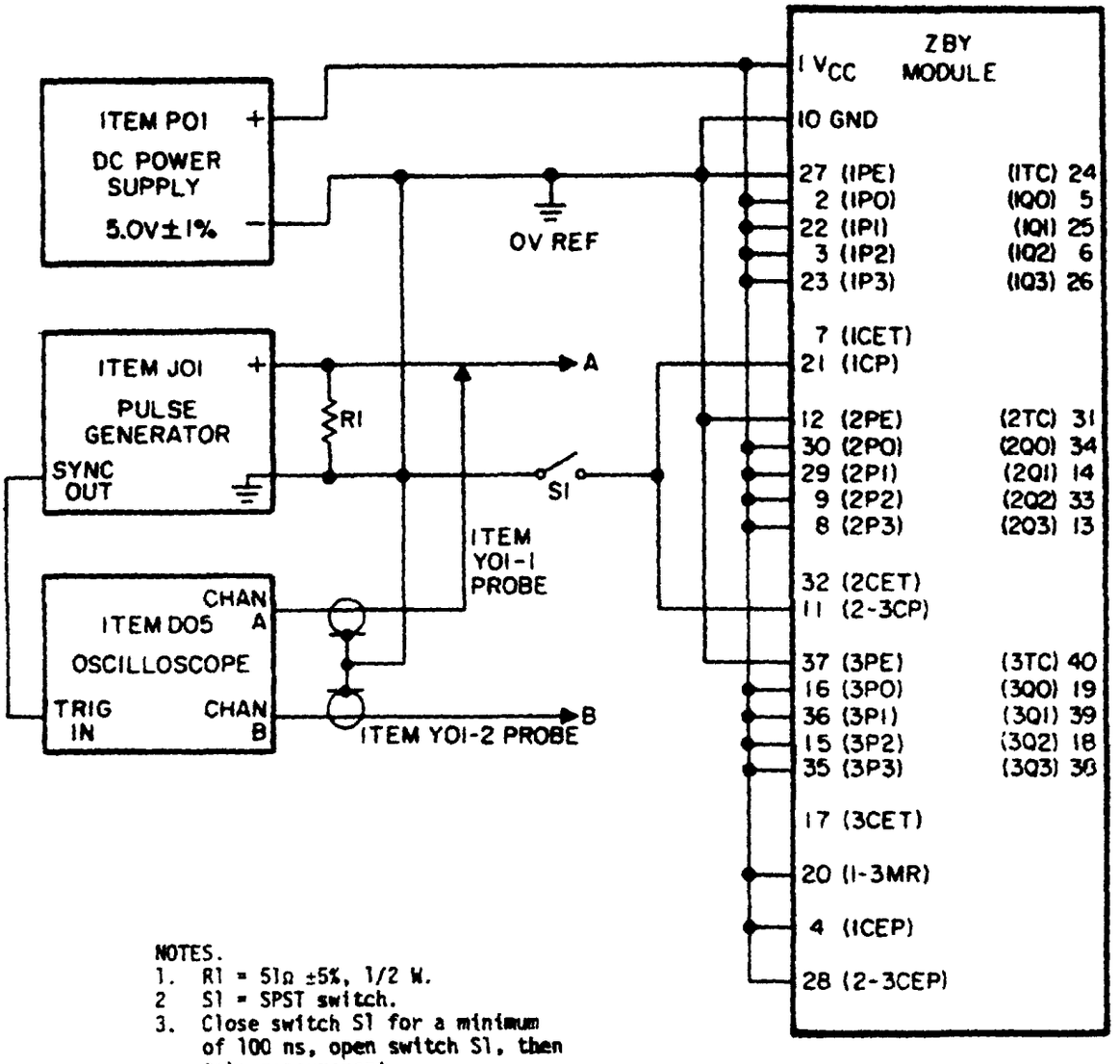
* FIGURE 20. Propagation delay time (t_{PHL}) test and minimum pulse width and recovery time check test circuit waveforms for table XXI.

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* TABLE XXII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
CC1	1CET to 1TC	Propagation delay time (t_{PHL}), C_L = 40 pF	7	24	10	64	ns	21,22
CC2	2CET to 2TC	Propagation delay time (t_{PHL}), C_L = 40 pF	32	31		64		
CC3	3CET to 3TC	Propagation delay time (t_{PHL}), C_L = 40 pF	17	40		64		
CC4	1CET to 1TC	Propagation delay time (t_{PLH}), C_L = 40 pF	7	24		59		
CC5	2CET to 2TC	Propagation delay time (t_{PLH}), C_L = 40 pF	32	31		59		
CC6	3CET to 3TC	Propagation delay time (t_{PLH}), C_L = 40 pF	17	40		59		
DD1	1CET to 1TC	Propagation delay time (t_{PHL}), C_L = 100 pF	7	24	15	70		
DD2	2CET to 2TC	Propagation delay time (t_{PHL}), C_L = 100 pF	32	31		70		
DD3	3CET to 3TC	Propagation delay time (t_{PHL}), C_L = 100 pF	17	40		70		
DD4	1CET to 1TC	Propagation delay time (t_{PLH}), C_L = 100 pF	7	24		65		
DD5	2CET to 2TC	Propagation delay time (t_{PLH}), C_L = 100 pF	32	31		65		
DD6	3CET to 3TC	Propagation delay time (t_{PLH}), C_L = 100 pF	17	40		65		

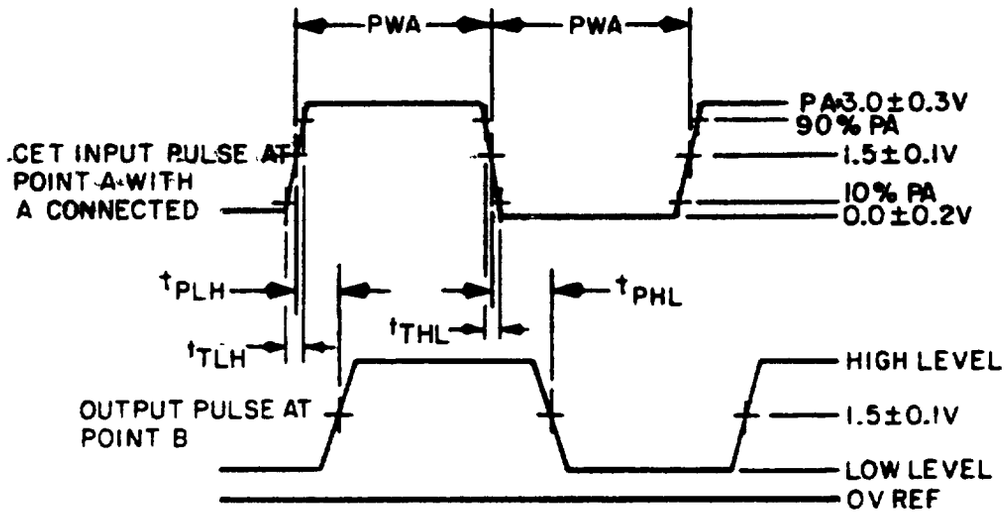
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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. Close switch S1 for a minimum of 100 ns, open switch S1, then take measurement.
4. For test condition CC, connect loads as specified on figure 10, load circuit 1 to pins 24, 31, and 40.
5. For test condition DD, connect loads as specified on figure 10, load circuit 2 to pins 24, 31, and 40.

FIGURE 21. Propagation delay time (t_{PHL} and t_{PLH}) test circuit for table XXII.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.

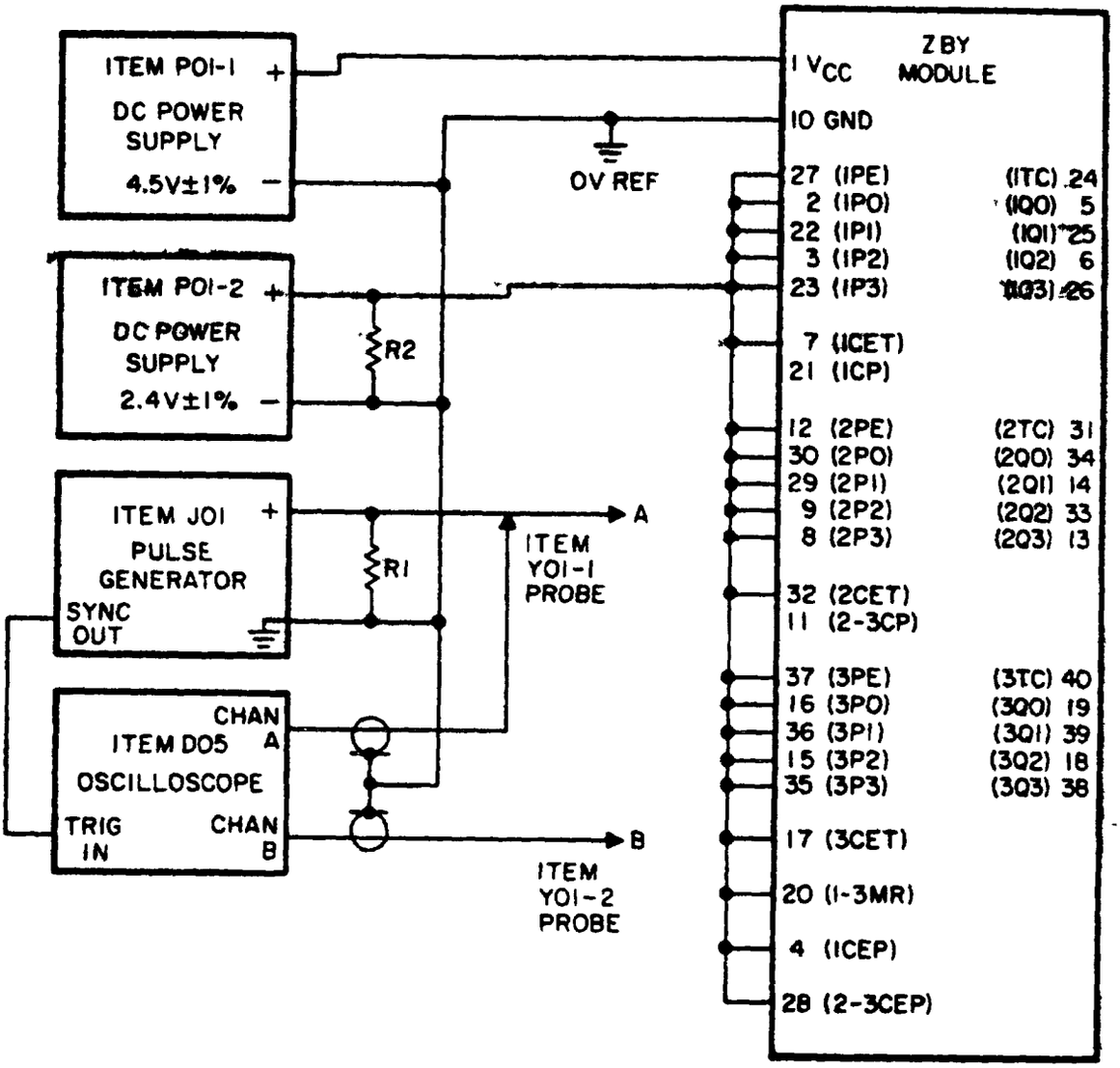
* FIGURE 22. Propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for table XXII.

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TABLE XXIII. Maximum counting frequency (minimum clock pulse width) check test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
EE1	PW1	Maximum counting frequency (minimum clock pulse width) check	21	26	0.97	1.07	µs	23,24
EE2	PW2	↓	21	26	↓	↓	↓	↓
EE3	PW1	↓	11	13	↓	↓	↓	↓
EE4	PW2	↓	↓	13	↓	↓	↓	↓
EE5	PW1	↓	↓	38	↓	↓	↓	↓
EE6	PW2	↓	↓	38	↓	↓	↓	↓

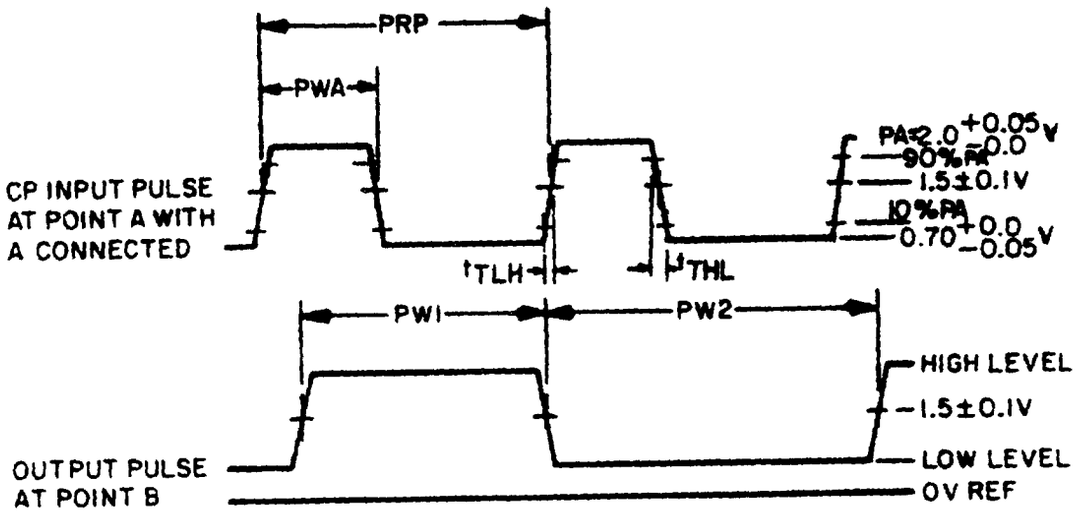
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Connect output loads as specified on figure 10, load circuit 1 to pins 5, 6, 13, 14, 18, 19, 24, 25, 26, 31, 33, 34, 38, 39, and 40.

* FIGURE 23. Maximum counting frequency (minimum clock pulse width) check test circuit for table XXIII.

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NOTES:
1. Input pulse parameters:
   PWA = 50 $^{+2}_{-0}$ ns.
   PRP = 125 $^{+5}_{-0}$ ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
2. All voltage levels are referenced to ground.
3 The time scales shown for pulses at points A and B are not the same.
4. Adjust 0.70 $^{+0}_{-0.05}$ V level at point A by placing item J01 in single pulse mode and adjusting the baseline voltage using item M02.

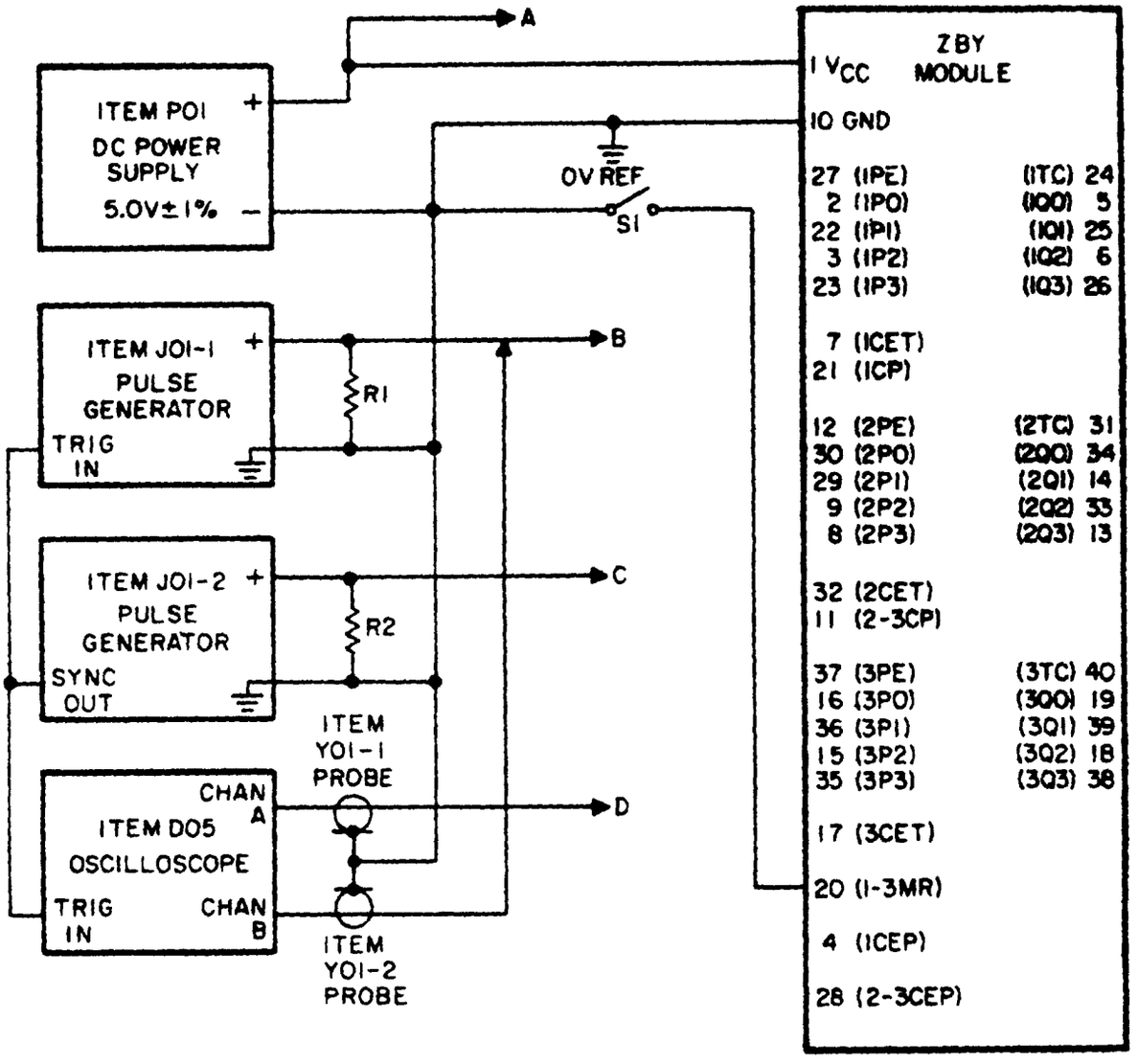
* FIGURE 24. Maximum counting frequency (minimum clock pulse width) check test circuit waveforms for table XXIII.

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TABLE XXIV. Setup and hold time (measuring output period) check test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	1CFT Setup and hold time (measuring output period) check	4	7	21	26	3.0	3.4	µs	25,26
FF2	2CET	28	32	11	13				
FF3	3CET	28	17	11	38				
FF4	1CEP	7	4	21	26				
FF5	2-3 CEP	32	28	11	13				
FF6	2-3CEP	17	28	11	38				

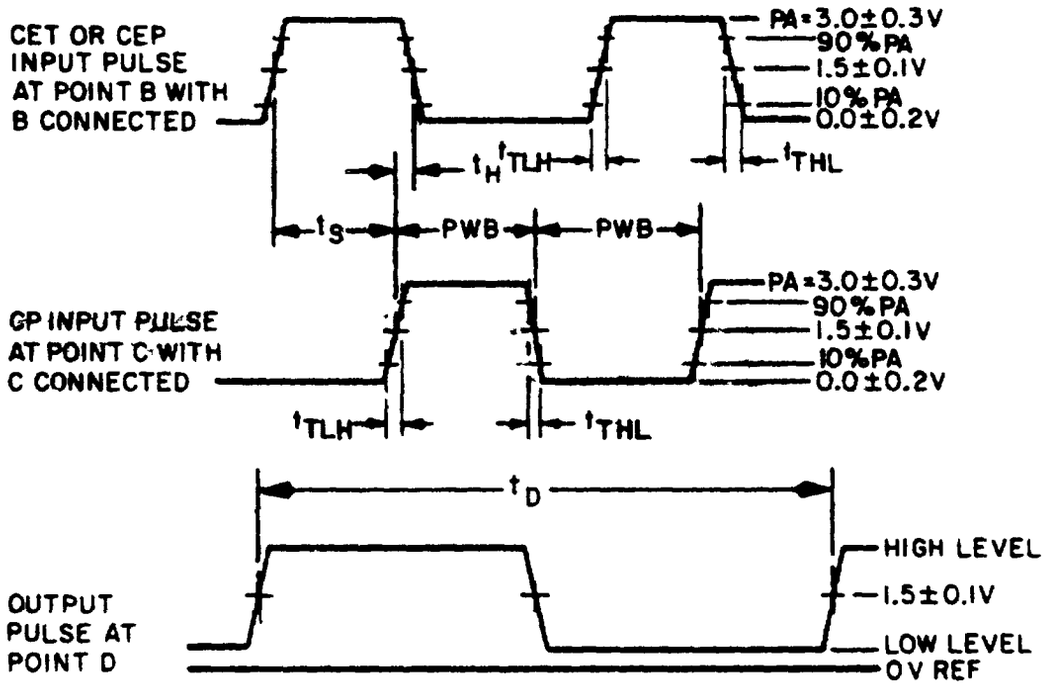
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. S1 = SPST switch.
3. After all circuit connections are made, close switch S1 for a minimum of 100 ns, then open it and make measurement.
4. Load outputs 26, 13, and 38 as specified on figure 10, load circuit 1.

FIGURE 25. Setup and hold time (measuring output period) check test circuit for table XXIV.

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NOTES:
1. Input pulse parameters:
   $t_S = 80\ ^{+4}_{-0}$ ns.
   $t_H = 0\ ^{+2}_{-0}$ ns.
   PWB = 100 ±5 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Output waveform at point D time scale is not the same as point B and C.
3. All voltage levels are referenced to ground.
4. t_D shall be as specified in table XXIV.

FIGURE 26. Setup and hold time (measuring output period) check test circuit waveforms for table XXIV.

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TABLE XXV. Setup and hold time (measuring t_{PLH} and t_{PHL}) check test.

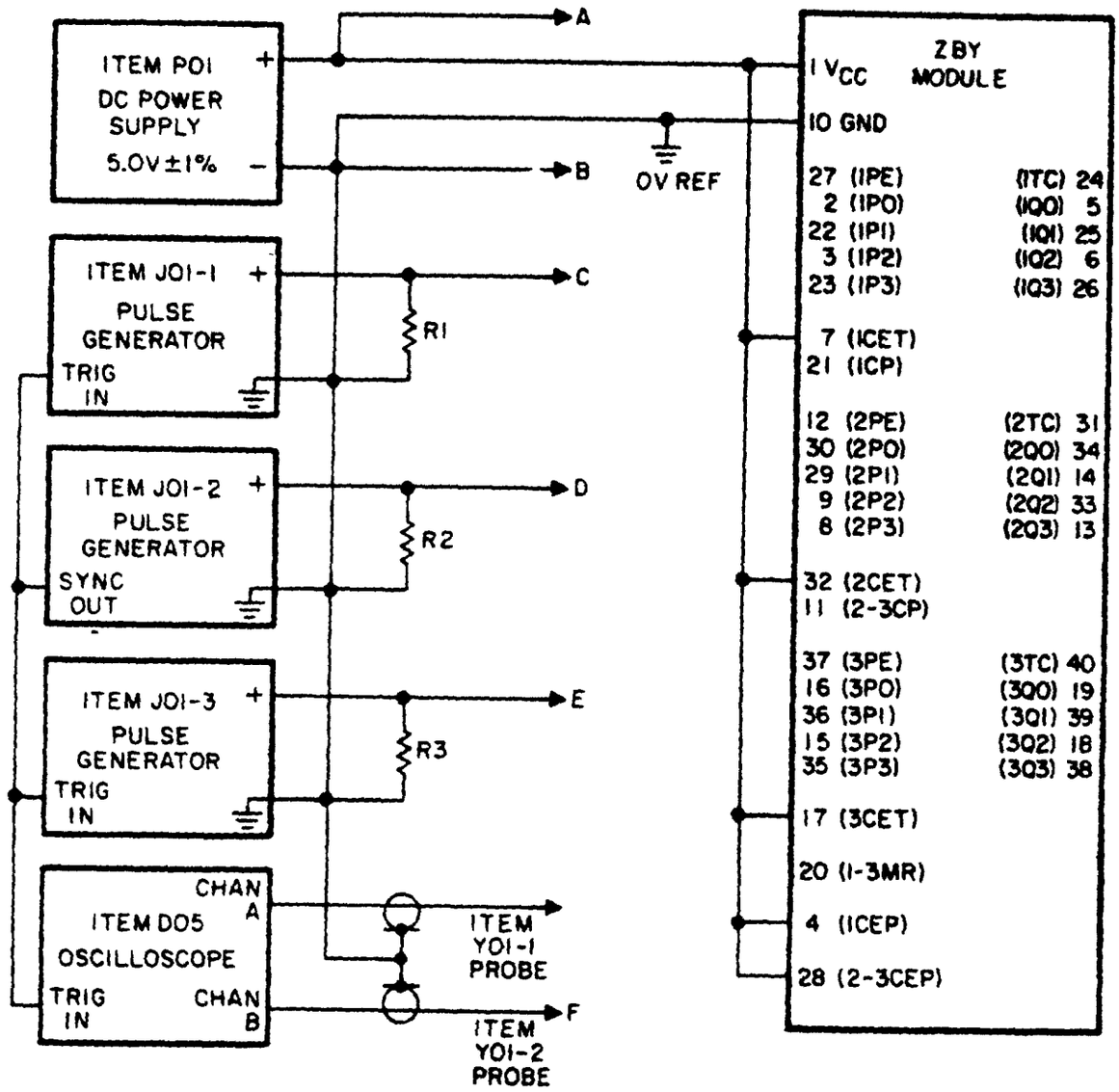
Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
GG1	1P0	Setup and hold time (measuring t_{PLH}), P high, check	20	27	2	21	Open	5	10	100	ns	27,28
GG2	1P1				22			25				
GG3	1P2				3			6				
GG4	1P3				23			26				
GG5	2P0			12	30	11		34				
GG6	2P1				29			14				
GG7	2P2				9			33				
GG8	2P3				8			13				
GG9	3P0			37	16			19				
GG10	3P1				36			39				
GG11	3P2				15			18				
GG12	3P3				35			38				
GG13	1PE	Setup and hold time (measuring t_{PLH}), PE, check	23	Open	27	21	20	26				27,29
GG.4	2PE	Setup and hold time (measuring t_{PLH}), PE, check	8	Open	12	11	20	13				27,29
GG15	3PE	Setup and hold time (measuring t_{PLH}), PE, check	35	Open	37	11	20	38				27,29

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TABLE XXV. Setup and hold time (measuring t_{PLH} and t_{PLH}) check test - Continued.

Test condition	Parameter		Connect						Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	F to pin	Min	Max		
GG16	1P0	Setup and hold time (measuring t_{PHL}), P low, check	Open	27	21	2	Open	5	10	100	ns	27,30
GG17	1P1					22		25				
GG18	1P2					3		6				
GG19	1P3					23		26				
GG20	2P0			12	11	30		34				
GG21	2P1					29		14				
GG22	2P2					9		33				
GG23	2P3					8		13				
GG24	3P0			37		16		19				
GG25	3P1					36		39				
GG26	3P2					15		18				
GG27	3P3					35		38				

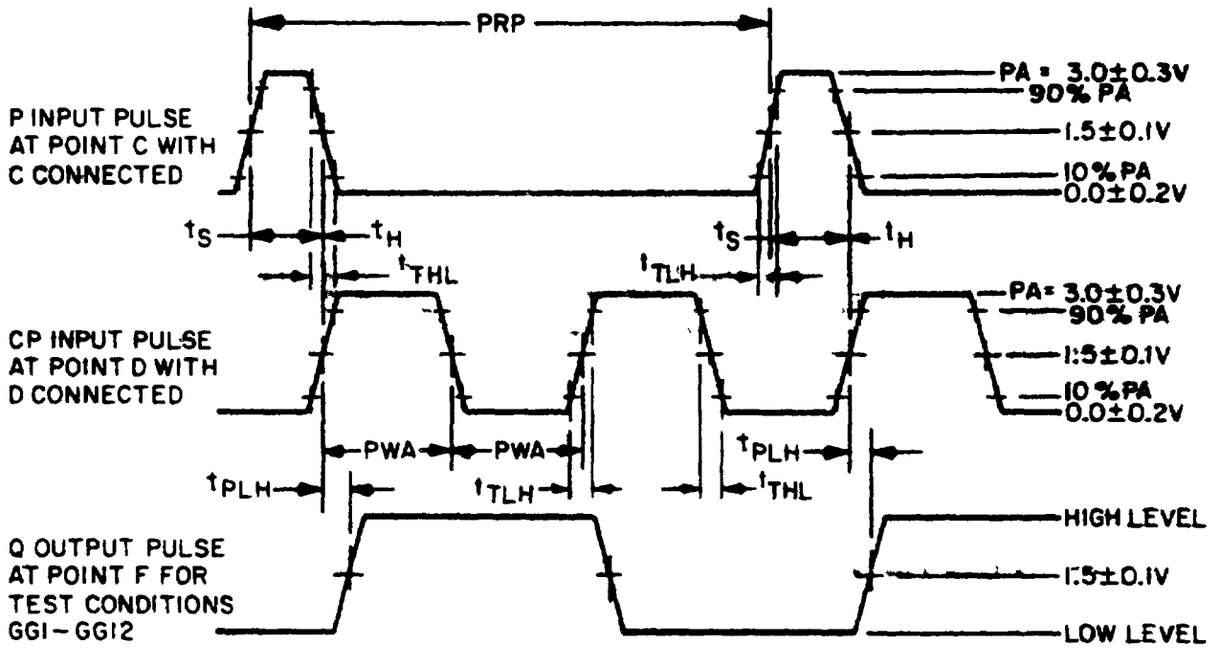
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NOTES:
1 R1-R3 = 51Ω ±5%, 1/2 W.
2 Load all outputs as specified on figure 10, load circuit 1.

FIGURE 27. Setup and hold time (measuring t_{PLH} and t_{PHL}) check test circuit for table XXV

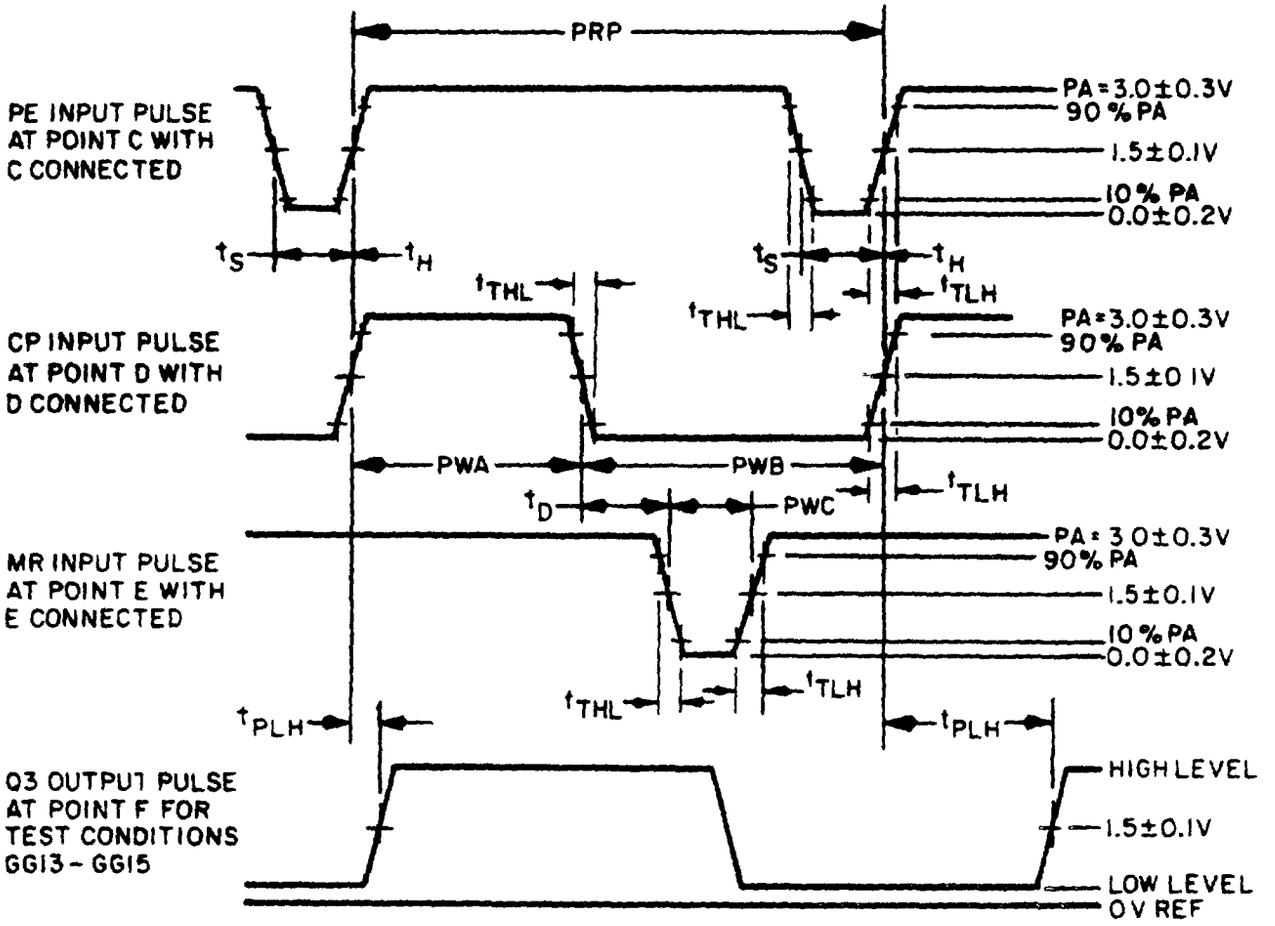
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NOTES:
1. Input pulse parameters:
   PWA = 250 ±25 ns.
   $t_S = 110\ ^{+5}_{-0}$ ns.
   $t_H = 0\ ^{+2}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PRP = 1,000 ±100 ns.
2. For test conditions GG1-GG12 place item J01-2 in the double pulse mode.
3. All voltage levels are referenced to ground.
4. t_{PLH} shall be as specified in table XXV.

* FIGURE 28. Setup and hold time (measuring t_{PLH}) check test circuit waveforms for test conditions GG1-GG12, table XXV.

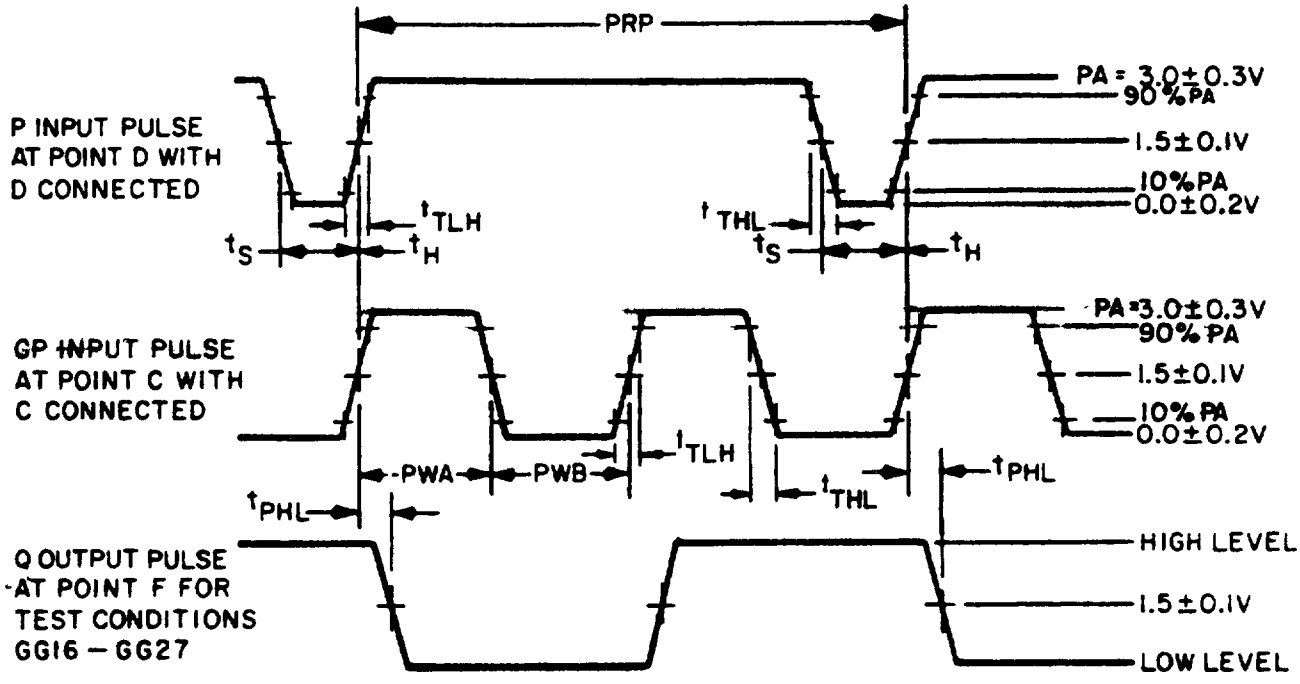
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NOTES.
1. Input pulse parameters:
   PWA = 400 ±40 ns.
   PWB = 600 ±40 ns.
   PWC = 100 ±10 ns.
   $t_S = 80 \,^{+4}_{-0}$ ns.
   $t_H = 0 \,^{+2}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   $t_D = 100 \pm 25$ ns.
   PRP = 1,000 ±100 ns.
2. All voltage levels are referenced to ground.
3. t_{PLH} shall be as specified in table XXV.

* FIGURE 29. Setup and hold time (measuring t_{PLH}) check test circuit waveforms for test conditions GG13-GG15, table XXV.

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NOTES:
1. Input pulse parameters:
   PWA = 250 ±25 ns.
   PWB = 250 ±25 ns.
   $t_S = 110 \,^{+5}_{-0}$ ns.
   $t_H = 0 \,^{+2}_{-0}$ ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   PRP = 1,000 ±100 ns.
2. For test conditions GG16-GG27 place item J01-1 in the double pulse mode.
3. All voltage levels are referenced to ground.
4. t_{PHL} shall be as specified in table XXV.

* FIGURE 30. Setup and hold time (measuring t_{PHL}) check test circuit waveforms for test conditions GG16-GG27, table XXV.

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TABLE XXVI. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	D to pin	E to pin	Min	Max		
HH1	1Q0 Output crosstalk, high level	2,7,17,32	3,8,9,15,16,22,23, 29,30,35,36	5	Open	0	400	mV	31,32
HH2	1Q1	7,17,22,32	2,3,8,9,15,16,23, 29,30,35,36	25					
HH3	1Q2	3,7,17,32	2,8,9,15,16,22,23, 29,30,35,36	6					
HH4	1Q3	7,17,23,32	2,3,8,9,15,16,22, 29,30,35,36	26					
HH5	1TC	2,3,7,17,22,23, 32	8,9,15,16,29,30, 35,36	24					
HH6	2Q0	7,17,30,32	2,3,8,9,15,16,22, 23,29,35,36	34					
HH7	2Q1	7,17,29,32	2,3,8,9,15,16,22, 23,30,35,36	14					
HH8	2Q2	7,9,17,32	2,3,8,15,16,22,23, 29,30,35,36	33					
HH9	2Q3	7,8,17,32	2,3,9,15,16,22,23, 29,30,35,36	13					
HH10	2TC	7,8,9,17,29,30, 32	2,3,15,16,22,23, 35,36	31					
HH11	3Q0	7,16,17,32	2,3,8,9,15,22,23, 29,30,35,36	19					
HH12	3Q1	7,17,32,36	2,3,8,9,15,16,22, 23,29,30,35	39					
HH13	3Q2	7,15,17,32	2,3,8,9,16,22,23, 29,30,35,36	18					

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TABLE XXVI. Output crosstalk (high-level and low-level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	D to pin	E to pin	Min	Max		
HH14	3Q3	Output crosstalk, high level	7,17,32,35	2,3,8,9,15,16,22,23,29,30,36	38	Open	0	400	mV	31,32
HH15	3TC	Output crosstalk, high level	7,15,16,17,32,35,36	2,3,8,9,22,23,29,30	40	Open		400		
JJ1	1Q0	Output crosstalk, low level	7,17,32	3,8,9,15,16,22,23,29,30,35,36	5	2		800		
JJ2	1Q1		7,17,32	2,3,8,9,15,16,23,29,30,35,36	25	22				
JJ3	1Q2		7,17,32	2,8,9,15,16,22,23,29,30,35,36	6	3				
JJ4	1Q3		7,17,32	2,3,8,9,15,16,22,29,30,35,36	26	23				
JJ5	1TC		17,32	2,3,8,9,15,16,22,23,29,30,35,36	24	7				
JJ6	2Q0		7,17,32	2,3,8,9,15,16,22,23,29,35,36	34	30				
JJ7	2Q1			2,3,8,9,15,16,22,23,30,35,36	14	29				
JJ8	2Q2			2,3,8,15,16,22,23,29,30,35,36	33	9				
JJ9	2Q3			2,3,9,15,16,22,23,29,30,35,36	13	8				
JJ10	2TC		7,17	2,3,8,9,15,16,22,23,29,30,35,36	31	32				

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TABLE XXVI. Output crosstalk (high-level and low level) test - Continued.

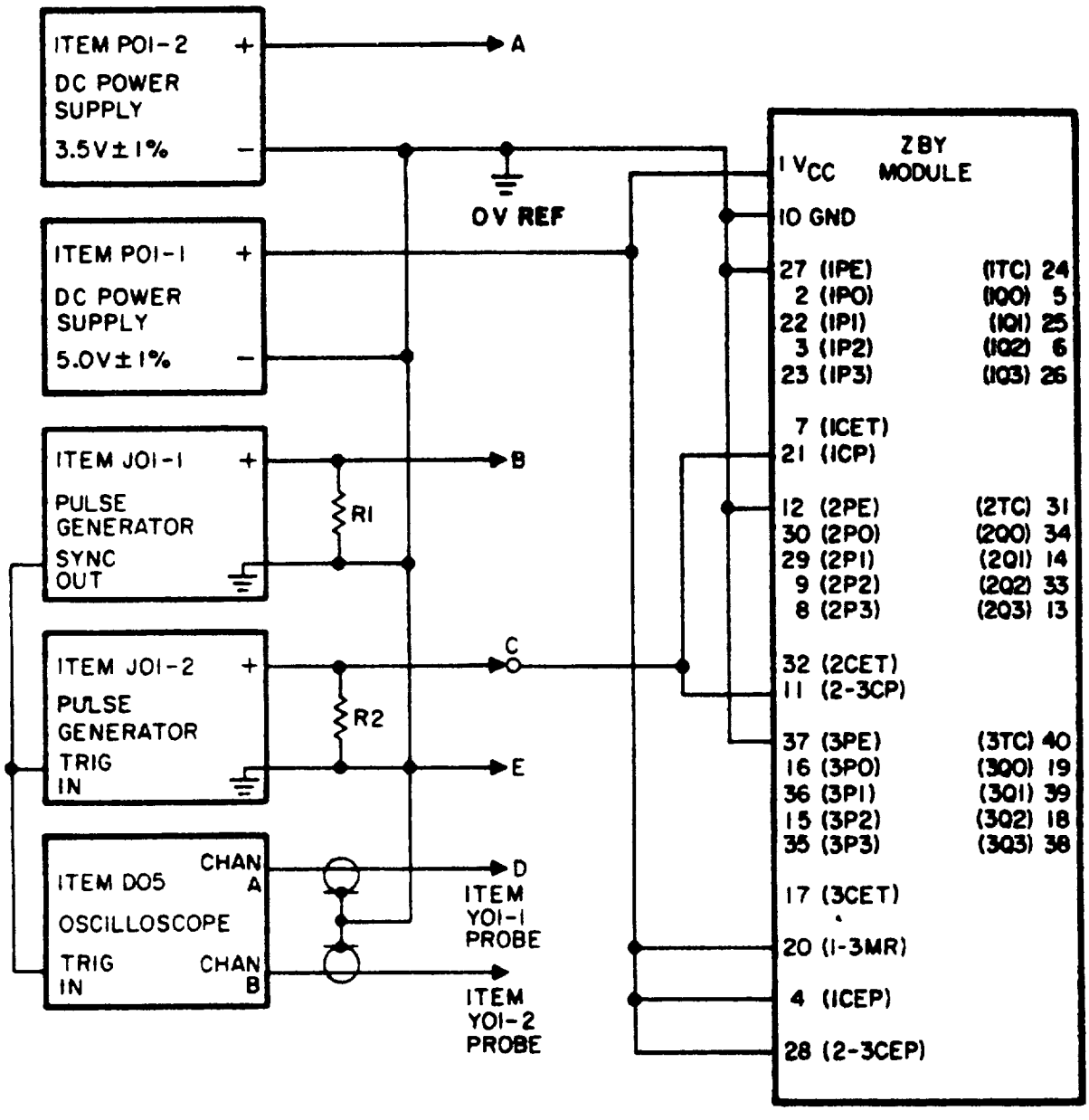
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	D to pin	E to pin	Min	Max		
JJ11	3Q0	Output crosstalk, low level	7,17,32	2,3,8,9,15,22,23, 29,30,35,36	19	16	0	800	mV	31,32
JJ12	3Q1			2,3,8,9,15,16,22, 23,29,30,35	39	36				
JJ13	3Q2			2,3,8,9,16,22,23, 29,30,35,36	18	15				
JJ14	3Q3			2,3,8,9,15,16,22, 23,29,30,36	38	35				
JJ15	3TC		7,32	2,3,8,9,15,16,22, 23,29,30,35,36	40	17				
JJ16	1Q0	Output crosstalk, low level, duration	7,17,32	3,8,9,15,16,22,23, 29,30,35,36	5	2		400		31,33
JJ17	1Q1		7,17,32	2,3,8,9,15,16,23, 29,30,35,36	25	22				
JJ18	1Q2		7,17,32	2,8,9,15,16,22,23, 29,30,35,36	6	3				
JJ19	1Q3		7,17,32	2,3,8,9,15,16,22, 29,30,35,36	26	23				
JJ20	1TC		17,32	2,3,8,9,15,16,22, 23,29,30,35,36	24	7				
JJ21	2Q0		7,17,32	2,3,8,9,15,16,22, 23,29,35,36	34	30				
JJ22	2Q1		7,17,32	2,3,8,9,15,16,22, 23,30,35,36	14	29				

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TABLE XXVI. Output crosstalk (high-level and low level) test - Continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	D to pin	E to pin	Min	Max		
JJ23	2Q2	Output crosstalk, low level, duration	7,17,32	2,3,8,15,16,22,23, 29,30,35,36	33	9	0	400	mV	31,33
JJ24	2Q3		7,17,32	2,3,9,15,16,22,23, 29,30,35,36	13	8				
JJ25	2TC		7,17	2,3,8,9,15,16,22, 23,29,30,35,36	31	32				
JJ26	3Q0		7,17,32	2,3,8,9,15,22,23, 29,30,35,36	19	16				
JJ27	3Q1			2,3,8,9,15,16,22, 23,29,30,35	39	36				
JJ28	3Q2			2,3,8,9,16,22,23, 29,30,35,36	18	15				
JJ29	3Q3			2,3,8,9,15,16,22, 23,29,30,36	38	35				
JJ30	3TC		7,32	2,3,8,9,15,16,22, 23,29,30,35,36	40	17				

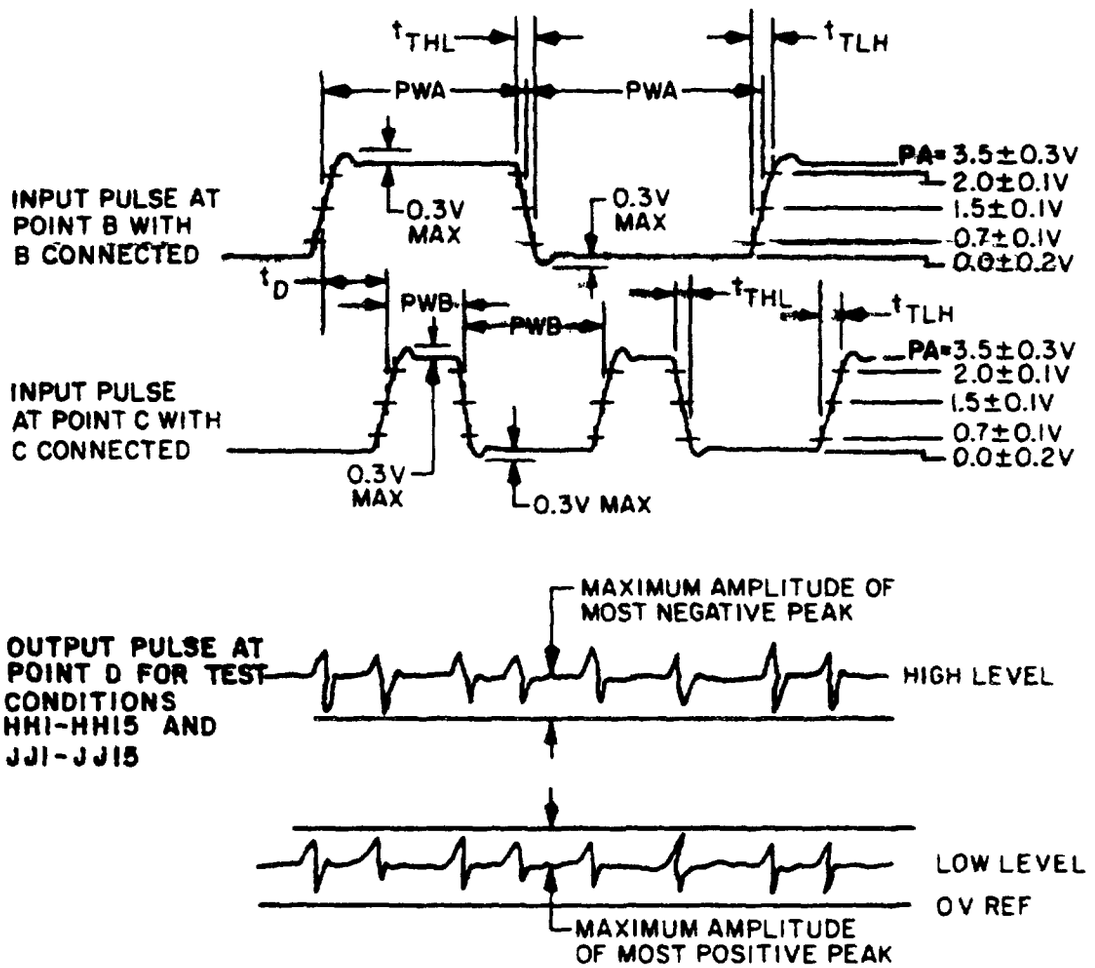
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load each output with a load as specified on figure 10, load circuit 1.

* FIGURE 31. Output crosstalk (high-level and low-level) test circuit for table XXVI.

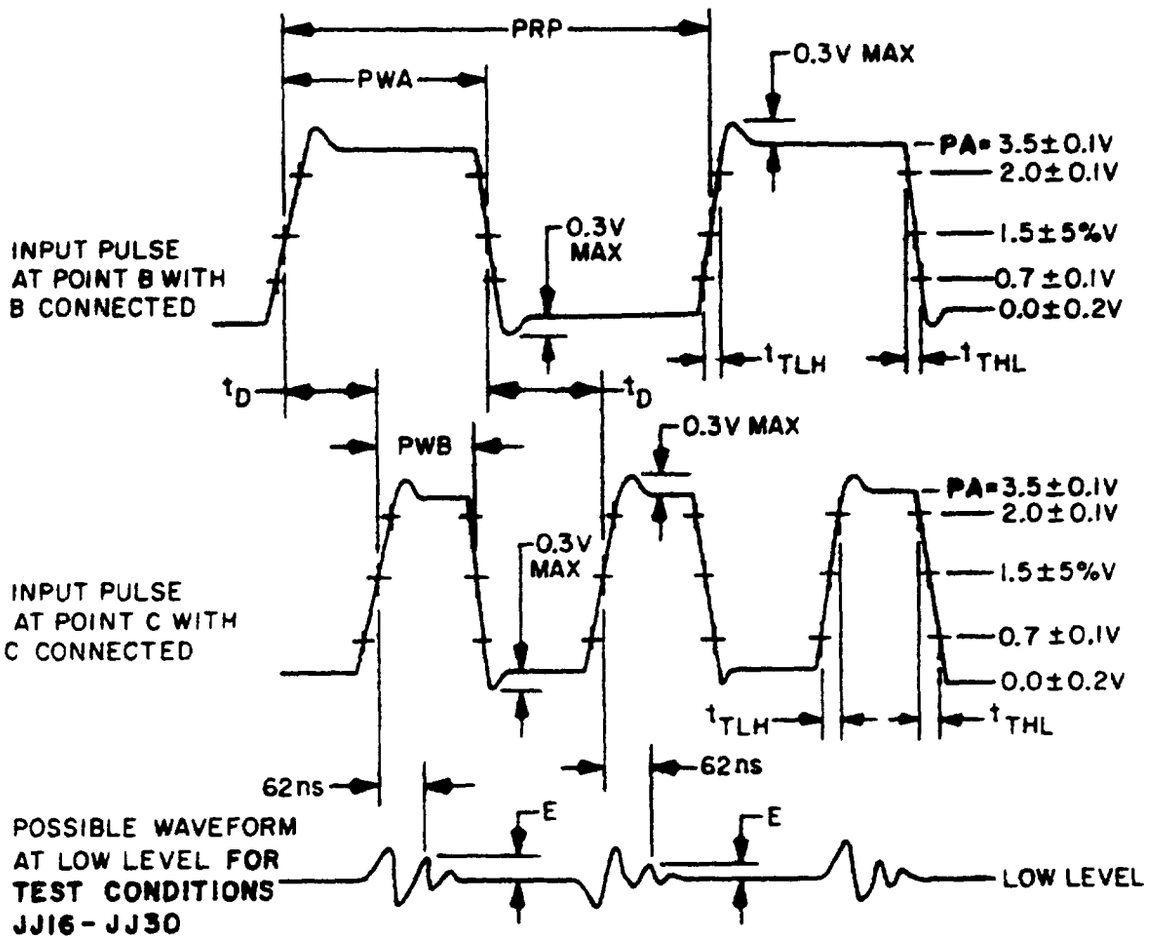
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NOTE: Input pulse parameters:
PWA = 500 ±50 ns.
PWB = 250 ±25 ns.
t_D = 250 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

* FIGURE 32. Output crosstalk (high-level and low-level) test circuit waveforms for table XXVI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
   PWB = 250 ±25 ns.
   t_{TLH} = 3.5 ±1.5 ns (verified at each input pin).
   t_{THL} = 3.5 ±1.5 ns (verified at each input pin).
   t_D = 250 ±50 ns.
2. E is measured from the low level to the most positive peak which occurs at least 62 ns after the positive going edge of the clock pulse.

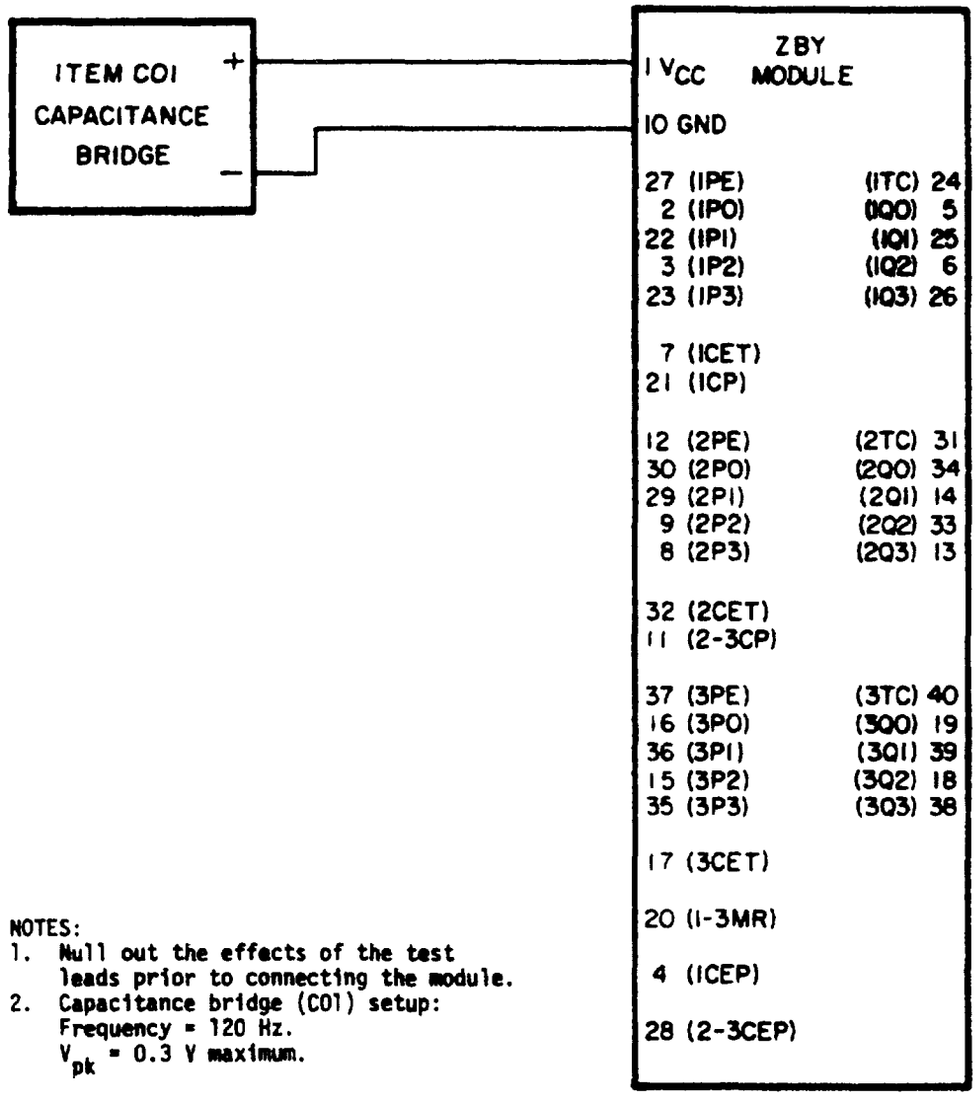
* FIGURE 33. Output crosstalk (low-level) test circuit waveforms for table XXVI.

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TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
KK	Power supply filter capacitance	1.6	3.0	µF	34

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Capacitance bridge (COI) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 34. Power supply filter capacitance test circuit for table XXVII.

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